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Graduation Project Report



Profound

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Abstract

This project, entitled **Profound**, aims to design and implement an AI-powered academic assistant platform that supports university professors in managing their teaching, grading, research, and administrative responsibilities. The study belongs to the field of educational technology and artificial intelligence, with a particular focus on Natural Language Processing (NLP) applications in grading and academic support.

The project is motivated by the growing challenges professors face, such as time-consuming grading processes, scattered teaching resources, difficulty in monitoring student performance, and ensuring academic integrity. To address these issues, the system integrates multiple modules within a centralized platform.

The proposed solution employs an **NLP-based grading model** for automatic evaluation of assignments and exams, a **materials generation and management module** for creating and sharing lecture notes and PDFs, and an **analytical dashboard** to monitor attendance, grades, and quizzes. Additional features include **plagiarism detection**, **exam generation tools**, and a **research organizer** to track deadlines and academic tasks. The project is developed using the **Agile methodology**, allowing iterative development, testing, and integration of features.

The results demonstrate that **Profound** can reduce professors' workload, improve grading consistency, and enhance overall academic productivity. The significance of this research lies in its contribution to educational technology by offering a unified system tailored specifically to the needs of professors. By streamlining repetitive academic tasks, the system enables educators to dedicate more time to teaching innovation and research development, ultimately improving the quality of higher education.

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List of Abbreviations

[illegible]

Chapter 1 Introduction

1.1 Background and Motivation

The role of university professors has become increasingly complex, as they are expected to balance teaching, research, grading, and administrative responsibilities. While technology has improved aspects of higher education, professors still face multiple challenges such as time-consuming grading processes, manual preparation of teaching materials, monitoring student performance, and ensuring academic integrity.

Traditional academic systems are fragmented and fail to provide an integrated environment that meets the needs of faculty members. This results in inefficiency, increased workload, and reduced focus on teaching innovation and research productivity.

The motivation behind the Profound project is to create a comprehensive AI-powered platform that supports professors in their daily academic tasks. By leveraging Natural Language Processing (NLP) and automation, the system aims to reduce workload, increase efficiency, and improve overall teaching and research experience.

1.2 Problem Statement

Current academic management systems lack integration and adaptability to the actual needs of university professors. Grading is often manual and time-consuming, manual preparation of teaching materials, and professors lack centralized tools to monitor class performance. Furthermore, plagiarism detection and exam generation require additional tools, which increase complexity and administrative burden.

Without a unified system, professors spend more time on repetitive administrative and grading tasks rather than focusing on teaching quality, student engagement, and research. Therefore, there is a critical need for a single intelligent platform that integrates grading automation, teaching material management, plagiarism detection, research organization, and analytics in one place.

1.3 Objectives

The main objectives of Profound are to:

- Develop a Grading NLP Model that automatically evaluates assignments and exams using customizable rubrics and provides feedback.
- Provide a Materials Generation and Management Module that allows professors to automatically generate lecture materials (slides, notes, and summaries) using AI.
- Integrate Plagiarism Detection to ensure academic integrity.
- Implement Exam Generation Tools that create questions based on Bloom's Taxonomy and provide variations to minimize cheating.
- Support Research Organization by helping professors track deadlines, papers, and projects.
- Introduce a Doctor Profile that allows professors to manage and showcase their academic background, research interests, publications, and teaching history within the system.
- Enhance professors' productivity and reduce administrative workload by consolidating all functions into one unified platform.

1.4 Project Scope and Limitations

Scope:

- The Profound system will be designed as a web-based academic assistant platform for professors. It will combine:
- An AI-powered grading system using NLP.
- A materials generation and management feature that allows professors to automatically generate lecture materials (slides, notes, and summaries) using AI.
- Modules for plagiarism detection, exam generation, and research organization.

Limitations:

- The plagiarism detection feature will rely on available online and institutional databases, which may limit accuracy.
- The grading NLP model will initially support only English-based assignments, with future work required for multilingual support.
- Development will be constrained by time, team size, and resources as this is a graduation project.

1.5 Project Planning

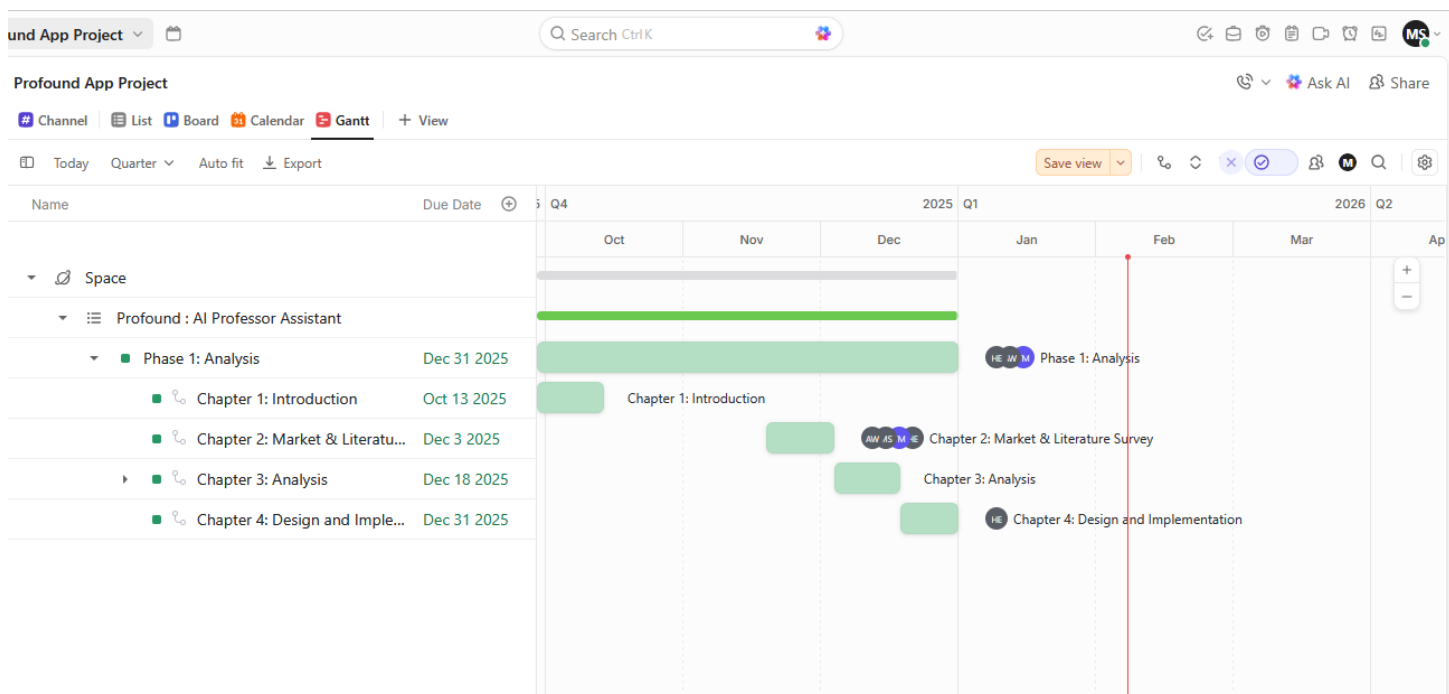


Figure 1. Project Plan

1.6 Project Methodology

To ensure flexibility and efficiency, the project will follow **Agile Methodology**. Agile is chosen because it allows iterative development and continuous improvement. Agile is suitable for projects with evolving requirements as it provides flexibility to integrate feedback, add features gradually, and ensure incremental delivery of functional modules.

Agile Process Steps:

1. Requirement Gathering: Collect user needs and system requirements.
2. Sprint Planning: Break the system into modules (grading NLP, materials generation, dashboard, etc.).
3. Iterative Development: Build and test features in short cycles.
4. Continuous Feedback: Review prototypes with the supervisor and adjust requirements.
5. Integration: Merge modules into a unified platform.

1.7 Project Report Outline

This outline provides a high-level structure for the project report and can be customized to fit the specific needs and requirements of the project. The purpose of each chapter is to provide a detailed and comprehensive overview of the various stages and aspects of the software development process, from the market and literature survey to the testing and implementation of the software system. The project report includes:

- Chapter 2: Market and Literature Survey
- Chapter 3: Software System Analysis
- Chapter 4: Profound Design and Implementation
- Chapter 5: Profound Testing and Evaluation
- Chapter 6: Conclusion and Future Work

Chapter 2 Market and Literature Survey

2.1 Marketing Plan

This section outlines the overall marketing approach of Profound, including objectives, performance metrics, revenue methods, and strategies to reach and engage professors and institutions.

2.1.1. Marketing Objectives

The marketing objectives of **Profound** focus on positioning the platform as a pioneering academic assistant that supports professors in teaching, grading, research, and administration. The main goals are to build strong brand recognition, expand adoption across universities, address the validated market need, and encourage the use of advanced features. These objectives follow the **SMART** principles:

- **Increase Awareness** - Grow awareness of Profound by 35% in the first year through online campaigns, webinars, and academic events. - **Rationale:** 76.5% of professors report that manual grading is their top challenge; highlighting AI solutions addresses this widespread concern.
- **User Adoption** - Reach 150 active professors within the first six months, starting with Egyptian universities. - **Rationale:** 62.9% of professors are “Very Interested” and 28.6% “Interested” in an AI-powered grading assistant, demonstrating strong market readiness.
- **Engagement** - Achieve 70% active monthly use in the first year by promoting key features such as NLP grading and plagiarism detection.
- **Market Expansion** - Partner with 3 Egyptian universities in the first year and expand regionally within two years.
- **Customer Satisfaction** - Achieve at least 85% satisfaction in professor feedback during the first year.

2.1.2 Key Performance Indicators (KPIs)

Shows how the success of the marketing objectives will be measured using specific metrics

- **User Growth & Acquisition**
 - Track new professor sign-ups and active users weekly/monthly.

- Insight: 91.5% of professors expressed interest in AI-assisted grading and exam generation, validating adoption targets.
- **User Engagement**
 - Monitor usage of core features (auto-grading, exam generation, content creation).
 - Track average time per session and number of tasks completed.
- **Market Expansion**
 - Monitor local and regional university partnerships.
 - Insight: 74.3% of professors want a centralized academic dashboard, indicating potential institutional adoption.
- **Brand Recognition**
 - Measure growth in social media engagement and university website traffic.
- **Customer Satisfaction**
 - Track overall satisfaction rate and recommendation percentage.
 - Measure user retention after 3 and 6 months.
- **Market Validation**
 - Collect survey feedback scores and positive reviews.
 - Monitor renewal and continued usage after pilot programs.
- **Revenue Growth**
 - Track paid subscriptions (individuals and institutions).
 - Measure revenue from premium features (analytics, plagiarism reports).

2.1.3 Profit Methods

Explains the different ways Profound generates revenue and the benefits of each method

- **Pay-Per-Use Services**
 - Users pay for specific AI-powered tools (e.g., AI grading, automatic exam generation, content creation, personalized learning recommendations).
 - Benefit: Flexible, scalable revenue based on usage.

- **Freemium Model**

- Offer a free basic version to attract new users; paid upgrades unlock advanced features like enhanced analytics and AI-assisted functionalities.
- Benefit: Encourages adoption while monetizing power users.

- **Subscription Plans**

- Monthly or yearly subscriptions for professors, researchers, or universities.
- Premium features include advanced analytics dashboards, plagiarism reports, AI insights, and personalized learning paths.
- Benefit: Provides stable, recurring revenue and strengthens user loyalty.

- **Institution Partnerships**

- Universities pay for campus-wide access or bulk licensing deals.
- Includes access to all platform tools, system integration, and support services.
- Benefit: Generates high-value B2B revenue and long-term contracts.

2.1.4 Marketing Strategies

Lists the main methods and approaches Profound will use to reach, attract, and retain professors and universities.

- **Digital Campaigns**

- **Goal:** Increase awareness among university professors and higher education institutions.
- **Strategy:** Targeted campaigns on LinkedIn, Twitter (X), and academic forums; organize live webinars and workshops.

- **Targeted Online Ads**

- **Goal:** Attract new professors and universities; increase platform sign-ups.
- **Strategy:** Paid ads on LinkedIn, Facebook, Google; direct ads to landing pages with free trial/demo.

- **Content Marketing**

- **Goal:** Establish Profound as a thought leader in AI for education.
- **Strategy:** Publish blog posts, whitepapers, podcasts, and case studies highlighting platform impact.

- **Email Marketing**

- **Goal:** Nurture leads, convert free users to paid plans, and maintain engagement.
- **Strategy:** Collect emails through sign-ups and content downloads; send personalized, value-driven emails and tutorials; automate sequences without spamming.

- **Influencer Marketing**

- **Goal:** Gain credibility and extend reach within the academic community.
- **Strategy:** Partner with respected professors, researchers, or educational content creators; start with micro-influencers for cost-effective engagement.

2.2 Related Work

Table 1. Similar Applications with unique advantages and current disadvantages

No.	System Name	Category	Advantages / Unique Value	Disadvantages
1	Gradescope	AI-Powered Grading	• AI-assisted grading with customizable rubrics • Supports multiple assignment formats (handwritten, code, diagrams) • Streamlines grading workflow for large classes • Provides analytics on student performance per question	• Requires manual rubric setup initially • Limited AI-generated feedback capabilities • Does not generate teaching materials or exams • No research organization features
2	Grammarly for Education	AI-Powered Feedback	• Real-time writing feedback and grammar correction • Plagiarism detection integrated • Supports institutional deployment • Helps improve student writing quality	• Focuses only on writing mechanics, not content evaluation • Cannot grade based on custom academic rubrics • No integration with course analytics or attendance • Limited to text-based assignments
3	Blackboard, Canvas (LMS)	Learning Management	• Widely adopted across universities worldwide	• Primarily course management tools • Lack advanced AI features (NLP grading,

		Systems	• Centralized platform for course content and delivery • Integration with assignment submission and gradebooks • Support third-party plugin integration	auto-material generation) • Limited professor-focused analytics • Heavy on student-facing features, light on professor workflow optimization
4	Google Classroom	Learning Management System	• Free and easy to use • Seamless integration with Google Workspace • Simple assignment distribution and collection • Real-time collaboration features	• Basic grading functionality only • No AI-powered grading or feedback • Limited analytics and reporting capabilities • Not designed for higher education complexity
5	Grammarly	Plagiarism Detection	• Leading plagiarism detection with large databases • Advanced similarity-matching algorithms • Strong role in maintaining academic integrity • Now includes AI writing detection (GPTZero integration)	• Operate as stand-alone services • Often require separate subscriptions • No integration with grading or performance dashboards • Cannot distinguish between proper citation and plagiarism in some cases
6	Quillionz	AI Question Generation	• Automatically generates questions from text content • Supports multiple question types (MCQ, true/false, short answer) • Saves time in exam preparation • Can align questions with learning objectives	• Question quality varies and requires manual review • Limited customization for advanced question types • Does not generate complete exams with variations • No integration with grading systems
7	ChatGPT, Claude (AI Assistants for Education)	AI Content Generation	• Generate lecture notes, summaries, and explanations • Create teaching materials and presentations • Assist with question generation	• General-purpose tools, not education-specific • Require manual integration into teaching workflow • No built-in rubrics or grading standards

			and feedback • Highly flexible and adaptable to various tasks	• Privacy concerns with student data • Output requires significant professor oversight
8	Gamma	AI Presentation Generators	• Automatically create visually appealing presentations • AI-driven content structuring • Save time in material preparation • Modern, professional templates	• Limited customization for academic content • Not integrated with course management • Cannot generate from existing course materials • No connection to grading or student analytics
9	Mendeley	Research Organization	• Strong citation and reference management • Collaboration features for research teams • Effective paper and deadline organization • Synchronization across devices	• Focused mainly on research productivity • Not tailored for professors' teaching needs • No linkage with grading or student monitoring • Do not help with teaching-research balance

2.3 Competitors' Features

Table 2. Key features across different applications

NO	Feature	Gradescope	Blackboard / Canvas	Google Classroom	Grammarly	Quillionz
1	AI Powered Grading	✓	⚠ Manual/gradebook tools only	⚠ Basic manual grading (no AI)	✗	✗
2	Plagiarism Detection	✓	✓	✓	✓	✗
3	Exam Generation	⚠ Question- bank based	✓	✓	✗	✓
4	Lecture Material Generation	✗	⚠ Manual upload only	⚠ Manual upload only	✗	⚠ Limited summaries
5	Research Organization	✗	✗	✗	✗	✗
6	Integration and Centralization	✓	✓	✓	✗	✗
7	Automation Level	Moderate	Low	Low	Moderate	Moderate

NO	Feature	Gamma	Mendeley	ChatGPT/Claude	Profound
1	AI Powered Grading	✗	✗	⚠ Can simulate grading via prompts but lacks rubric evaluation	✓ Advanced NLP grading model with customizable rubrics and automatic feedback
2	Plagiarism Detection	✗	✗	⚠ Can identify reused text (not reliable academically)	✓ Integrated plagiarism detection using institutional and web databases
3	Exam Generation	✗	✗	⚠ Can generate questions manually through prompts	✓ AI exam generation aligned with Bloom's Taxonomy and automatic question variation
4	Lecture Material Generation	✓	✗	✓	✓ AI generation of lecture notes, slides, and summaries customized to the course
5	Research Organization	✗	✓	✓	✓ Rewrites research in professional academic tone Get research Personalized to user's interests
6	Integration and Centralization	✗	✗	✗	✓ Unified AI-powered platform combining all academic functions
7	Automation Level	High	Moderate	High	Very High (academic workflow automation)

2.4 SWOT Analysis

1. Strengths

1. **Comprehensive All-in-One Platform:**

Integrates AI-based grading, plagiarism detection, material generation, exam creation, and Excel data analysis — all designed specifically for university professors.

2. **Intelligent Excel Data Analytics:**

Automatically processes uploaded Excel sheets (attendance, submissions, grades, etc.) to provide actionable insights such as identifying at-risk students, tracking participation, and analyzing performance trends.

3. **Academic Profile System (Professor Hub):**

Each professor has a personal academic profile showcasing their research, courses taught, graduation projects supervised, and academic interests — creating a professional presence similar to LinkedIn but specialized for academia.

4. **AI-Powered Automation and NLP Feedback:**

Automates grading and feedback generation with high accuracy and consistency using Natural Language Processing and customizable rubrics.

5. **Strong Focus on Professor Needs:**

Unlike traditional LMS tools, Profound is designed *for professors first*, addressing real academic pain points such as grading overload, scattered materials, and lack of organized academic visibility.

6. **Ease of Use and Scalability:**

Modular design allows gradual feature adoption — from grading automation to analytics and profiles — making it adaptable for individuals or institutions.

7. **Data-Driven Decision Support:**

Enables professors to make informed academic decisions based on actual performance data rather than manual tracking or intuition.

2. Weaknesses

1. **Limited Language Support (English Only):**

NLP-based features currently work only for English text, reducing accessibility for professors teaching in Arabic or other languages.

2. Dependence on Uploaded Data Quality:

Analytics accuracy depends on the correctness and consistency of Excel sheet data provided by professors.

3. Privacy and Data Sensitivity:

Storing academic profiles, grades, and research data requires strong privacy control, encryption, and compliance with institutional data policies.

4. Development Constraints:

Being a graduation project, the system may initially lack advanced infrastructure (servers, security layers, scalability) available in commercial tools.

5. Adoption Resistance from Non-Tech Users:

Professors who are less comfortable with digital tools might face a learning curve when uploading data or managing their profiles.

3. Opportunities**1. Academic Networking and Collaboration:**

The professor profiles can evolve into a mini-academic network, connecting educators for joint research, mentorship, or co-supervision — filling a gap that LinkedIn doesn't serve well.

2. Institutional Integration:

Universities can use Profound as an internal academic ecosystem, linking profiles, grading, and analytics in one system.

3. Regional and International Expansion:

Supporting Arabic and other languages can make Profound a leading academic tool in Egypt, MENA, and global markets.

4. Predictive Student Analytics:

Expanding the Excel-based analytics to include predictive modeling can help universities detect academic risks early and improve retention rates.

5. University Branding and Portfolios:

Institutions could use aggregated professor profiles to showcase their academic staff and research output on official websites.

6. Subscription and Licensing Models:

Freemium and institutional subscription options can provide sustainable funding and scalability beyond the academic phase.

7. **Collaboration with Research Databases:**

Integration with platforms like Google Scholar, Scopus, and ResearchGate can automatically update professors' publications and citations.

4. **Threats**

1. **Competition from Academic Networks and LMS Tools:**

Existing platforms like ResearchGate, Google Scholar, and Blackboard may introduce similar features that overlap with Profound's vision.

2. **Data Security Regulations:**

Handling academic and personal data will require compliance with strict university or government data protection policies.

3. **Rapid Technological Advancement:**

Continuous evolution in AI and educational technology may require frequent updates to remain relevant and competitive.

4. **Adoption Resistance:**

Some professors might distrust AI-generated grades or analytics, preferring traditional manual systems.

5. **Institutional Bureaucracy:**

University approval for using third-party academic systems may delay or restrict large-scale deployment.

2.5 Key Learnings and Opportunities

2.5.1 Key Learnings

Throughout the development and market research of **Profound**, several important insights emerged regarding the use of artificial intelligence and automation in higher education:

1. **Professors Face Increasing Digital and Administrative Overload:**

Survey responses and interviews revealed that professors spend a significant portion of their time on repetitive tasks such as grading, attendance tracking, and report generation. This validated the need for a centralized AI-driven tool that saves time and enhances efficiency.

2. **Existing Systems Focus Mainly on Students, Not Professors:**

Current educational platforms like Google Classroom, Blackboard, and Canvas

primarily cater to student learning management rather than professor productivity.

This gap confirmed the necessity for a **professor-centered platform** that integrates teaching, grading, and research tools in one space.

3. **AI Can Improve Fairness and Consistency in Grading:**

NLP-based evaluation reduces human bias and subjectivity, providing fairer and more standardized grading while still allowing professors to customize rubrics and feedback styles.

4. **Data Analytics from Existing Sources Is Underutilized:**

Professors already maintain valuable data in Excel sheets (attendance, grades, performance), but this data is rarely analyzed for insights. Profound's ability to process these sheets demonstrated that meaningful predictions—such as identifying at-risk students—can be generated with minimal effort.

5. **Academic Profiles Encourage Visibility and Collaboration:**

The introduction of professor profiles highlighted the potential for **academic networking** and professional exposure. Professors appreciated the idea of having a space to display their research, supervised projects, and academic achievements, which fosters collaboration opportunities within and across institutions.

6. **Ease of Use and Integration Are Key Success Factors:**

Professors value tools that integrate easily into their existing workflow. Systems requiring minimal data re-entry (e.g., uploading existing Excel sheets) are far more likely to be adopted than those needing full migration.

2.5.2 Opportunities for Growth and Expansion

Building on these learnings, several opportunities were identified for future development and scalability of Profound:

1. **Institutional Collaboration and Integration:**

Partnering with universities to integrate Profound with existing LMS systems (e.g., Google Classroom, Blackboard) can automate data synchronization and improve adoption rates.

2. **Multilingual Expansion:**

Extending NLP support to Arabic and other languages would broaden Profound's user base across Egypt, the Middle East, and non-English-speaking regions worldwide.

3. **AI-Driven Predictive Analytics:**

Enhancing the Excel analysis module with predictive models can forecast student performance trends, enabling early intervention for those at risk of failure.

4. **Research and Publication Integration:**

Connecting the professor profiles with platforms like **Google Scholar**, **Scopus**, and **ResearchGate** would automatically update research output and strengthen academic networking.

5. **Institutional Analytics and Benchmarking:**

Aggregating anonymized data from multiple professors can provide universities with insights on department-level performance, teaching effectiveness, and student success rates.

6. **Mobile and Cross-Platform Access:**

Developing a mobile-friendly version or dedicated mobile app can increase accessibility and daily engagement for professors on the go.

7. **Commercialization and Sustainability:**

The system can evolve into a **SaaS (Software as a Service)** model with subscription-based plans for universities and premium users, ensuring sustainability beyond the graduation phase.

8. **AI Personalization:**

Implementing personalized recommendations (e.g., “suggested research topics,” “grading patterns,” or “student engagement alerts”) can make Profound an intelligent assistant rather than a passive tool.

Chapter 3 Profound Analysis

3.1 Stakeholder

This section identifies stakeholders as individuals or groups who influence or are affected by profound platform. They are divided into primary stakeholders (directly involved in system use and development) and secondary stakeholders (providing guidance and support).

Primary Stakeholders

- **Professors and Educators**
 - Core users of the system responsible for creating and managing courses, generating AI-assisted exams, and grading assignments.
 - They use analytics dashboards to monitor student progress and adjust teaching strategies based on insights.
 - Their feedback drives system improvement and feature updates.
- **Students**
 - Students benefit indirectly from the AI Academic Management System through improved teaching quality and faster grading enabled by AI-powered tools used by professors.
 - They receive more accurate and timely results, helping them track their academic progress effectively.
 - The system also enhances academic integrity by minimizing human errors and detecting plagiarism before results are released.
- **System Administrators**
 - Manage system operations, user accounts, access levels, and data security.
 - Ensure that the platform remains functional, secure, and up to date.
 - Provide technical support to both professors and students

- **Project Development Team**

- **Project Manager:** Oversees the development process, manages milestones, and ensures timely delivery.
- **Developers:** Implement system architecture, AI functionalities, and backend services.
- **UI/UX Designers:** Ensure intuitive and engaging user experience for professors and students.
- **AI Specialists/Data Scientists:** Build intelligent grading, recommendation, and prediction models.
- **QA Engineers:** Test system performance, fix bugs, and maintain quality before deployment.

3.2 User persona

This section describes the main types of users who interact with the Profound platform. Each persona represents a group of users with similar goals and needs. These people help the development team understand user expectations and design features that truly support their work.



Figure 2. User Persona 1

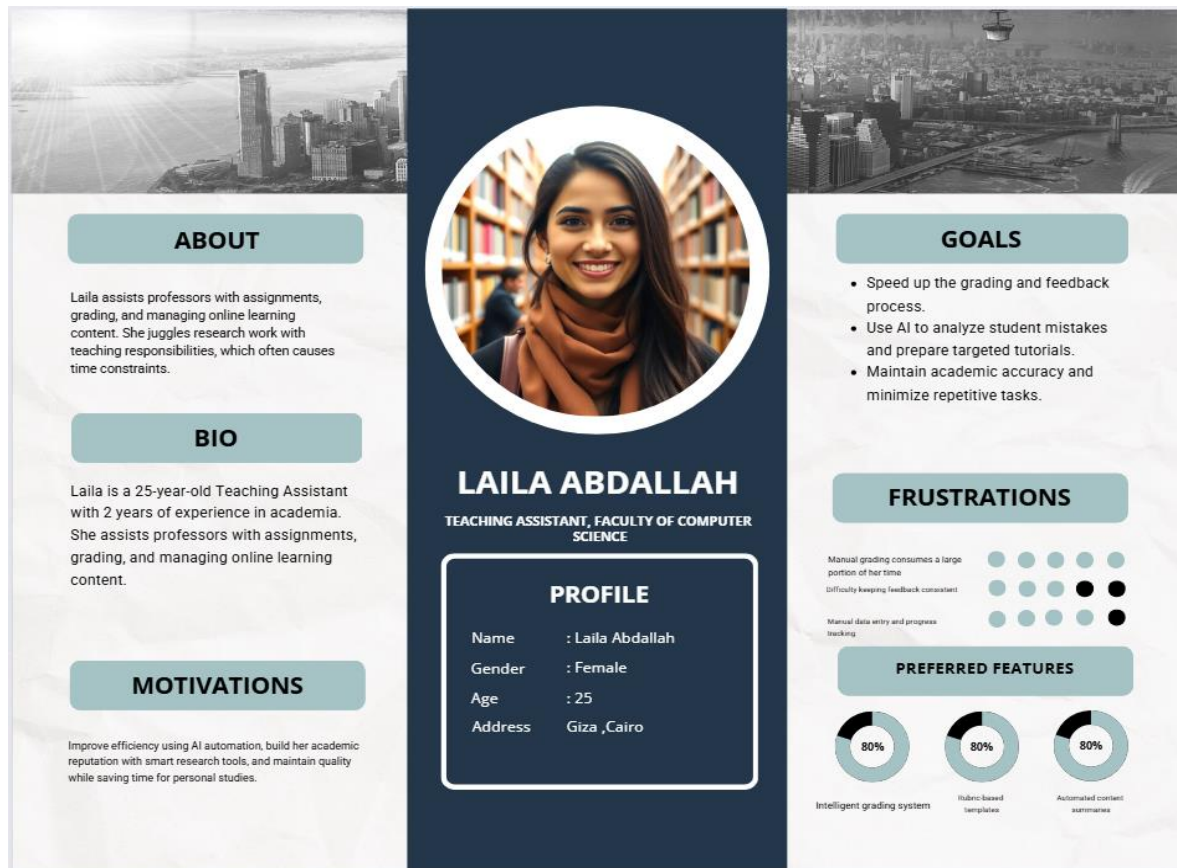


Figure 3. User Persona 2

3.3 Functional Requirements

Functional requirements define what the Profound system must perform to achieve its goals and meet the needs of professors, teaching assistants, and students.

1. User and Access Management

- Registration and Authentication
 - Allow professors to register through university email or institutional credentials.
- Profile Management
 - Enable users to update personal and academic details.

2. Course and Content Management

- Content Creation Tool
 - Allow professors to generate lecture notes, summaries, and learning materials using AI.

- Support multimedia (videos, images, slides) for interactive learning.
- Provide a PowerPoint presentation maker to create structured, ready-to-use decks.
- Student Engagement and Feedback Tools
 - Integrate interactive polls, quick quizzes, and AI-based feedback analysis to measure engagement and sentiment.
- Advanced AI Assistant for Lecturers
 - Assist in overall course preparation, suggesting step-by-step guidance for lecture planning.
 - Convert spoken explanations into slides automatically in real-time (speech-to-slide feature).

3. Smart Exam and Question Generation

- AI Question Bank Generator
 - Creates multiple-choice, essay, or coding questions based on uploaded lecture materials or selected topics.
 - Automatically classifies generated questions by cognitive level (e.g., Remember, Apply, Analyze).
 - Generates multiple exam versions to reduce cheating while maintaining question difficulty balance.

4. Analytics and Reporting

- Dashboard for Professors
 - Display student performance, grading summaries, and class progress visually.
 - Allow custom report views (weekly, monthly, or semester-based).
 - Provide AI-driven teaching insights to identify weak topics and improve course delivery
 - Allow exporting analytics and reports in PDF or Excel formats.
- Department-Level Analytics

- Provide insights into faculty performance, course efficiency, and student outcomes.
- Predictive Analysis
 - Use AI to detect struggling students early and recommend intervention or extra support.
- Attendance and Engagement Reports
 - Provide automated attendance tracking and visual analytics for class participation.
- Quality Assurance Reports
 - Generate academic performance and course evaluation summaries for accreditation or internal review.

5. Evaluation and Grading

- AI-Based Grading System
 - Automatically evaluate assignments and quizzes with consistent accuracy.
 - Provide plagiarism detection and automated feedback.
- Manual Override
 - Professors can review and adjust AI grades manually if needed.
- Intelligent Grading and Feedback
 - Analyze common student errors across exams to guide teaching improvements.
 - Support rubric-based grading templates to ensure fairness and consistency.

6. Research Support

- Robotics and Machine Learning Integration
 - Enable automation and high-throughput research tasks using AI-driven robotics and intelligent data synthesis tools.

3.4 Non-Functional Requirements

1. Usability

The system is targeted at professors who may not have advanced technical expertise; therefore, the user interface should be intuitive, visually consistent, and require minimal training. Each major function—such as uploading assignments, generating exams, or viewing analytics—should be accessible within two clicks from the dashboard. Clear guidance messages, tooltips, and help documentation will be embedded to support new users and reduce the learning curve.

2. Performance

The system must deliver real-time responses with minimal delay to maintain a seamless user experience. All dashboard visualizations, reports, and grading results should load within a few seconds under normal server load. The NLP grading engine is expected to analyze and return grades for typical text assignments within five seconds per submission. Caching mechanisms and optimized database queries will be used to prevent bottlenecks as data grows. Overall performance must remain stable even when processing multiple concurrent grading requests.

3. Security

Security is essential to preserve the confidentiality, integrity, and availability of academic data. Profound will ensure that access to the system is restricted to authorized users only, following clearly defined roles and permissions for professors, students, and administrators. Sensitive information such as user credentials, grades, and research data will be protected through secure storage and controlled access. The platform will comply with institutional data protection guidelines, and user activities will be monitored to prevent unauthorized use or disclosure.

4. Compatibility

5. Scalability

The platform should be designed to expand easily as the number of users and data grows. The architecture will allow additional servers or service instances to be added without major redesign. The database structure and AI components must support scaling to accommodate new universities,

professors, and integrated features. This scalability ensures that Profound remains sustainable beyond initial deployment.

6.. Maintainability

The system must be easy to maintain and evolve. Modular design principles will be applied so that updates to one feature (e.g., plagiarism detection) do not disrupt others. All components will follow consistent naming conventions and documentation standards to simplify troubleshooting and future enhancement. Regular maintenance activities such as log reviews, updates, and backups should be easily executable by administrators.

7. Reliability

As an academic tool used for daily operations, Profound must operate consistently and be available whenever needed. The system should provide high reliability through error handling, recovery procedures, and automated data backups. All essential data—such as grades, rubrics, and teaching materials—must be protected from accidental loss or corruption. Planned maintenance should not interrupt critical grading or reporting activities

3.5 User Stories

Table 3. Automated Assignment Grading User Story

User Story ID	US-1
User Story Name	Automated Assignment Grading
Actors	Professor
User Story	As a professor, I want to automatically grade assignments using an AI model so that I can reduce the time spent on manual grading and ensure consistent evaluation across students.
Acceptance Criteria	Given that a professor uploads student submissions, when the grading process is initiated, then the system should analyze the content using the NLP model, generate grades and feedback based on the provided rubric, and display results in the grading dashboard for review and confirmation.

Table 4. AI-Generated Lecture Materials User Story

User Story ID	US-2
User Story Name	AI-Generated Lecture Materials
Actors	Professor
User Story	As a professor, I want to automatically generate lecture notes, slides, and summaries using AI so that I can prepare my teaching materials more efficiently and maintain high-quality educational content.
Acceptance Criteria	Given that a professor provides a topic or uploads reference materials, when the content generation process is triggered, then the system should produce well-structured lecture materials that can be reviewed, edited, and exported in multiple formats.

Table 5. Plagiarism Detection User Story

User Story ID	US-3
User Story Name	Plagiarism Detection
Actors	Professor
User Story	As a professor, I want to detect plagiarism in student submissions so that I can uphold academic integrity and identify copied or unoriginal work.
Acceptance Criteria	Given that a professor uploads student files for evaluation, when the plagiarism detection module runs, then the system should compare the text with available databases and display a similarity report showing percentage matches and highlighted sources.

Table 6. Exam and Question Generation User Story

User Story ID	US-4
User Story Name	Exam and Question Generation
Actors	Professor
User Story	As a professor, I want to automatically generate exam questions based on Bloom's Taxonomy so that I can create fair and diverse assessments for my students.
Acceptance Criteria	Given that a professor selects a subject or uploads course materials, when the exam generation feature is activated, then the system should generate categorized questions by difficulty level, allowing the professor to review, edit, and export them in preferred formats.

Table 7. Student Performance Analytics User Story

User Story ID	US-5
User Story Name	Student Performance Analytics
Actors	Professor
User Story	As a professor, I want to analyze student performance and attendance data so that I can identify learning trends and support at-risk students effectively.
Acceptance Criteria	Given that a professor uploads grade or attendance sheets, when the analytics dashboard is accessed, then the system should display interactive charts and insights summarizing student performance, class averages, and participation rates.

Table 8. Professor Academic Profile User Story

User Story ID	US-6
User Story Name	Professor Academic Profile
Actors	Professor
User Story	As a professor, I want to maintain a professional profile with my research, publications, and teaching activities so that I can manage and showcase my academic contributions.
Acceptance Criteria	Given that a professor accesses their profile section, when they update details such as publications, courses, or research interests, then the system should save and display the information securely and consistently across all modules.

Table 9. Rubric Management User Story

User Story ID	US-7
User Story Name	Rubric Management
Actors	Professor
User Story	As a professor, I want to create and customize grading rubrics so that I can align automated grading with my specific evaluation standards.
Acceptance Criteria	Given that a professor opens the rubric configuration interface, when they define evaluation criteria and assign weights, then the system should store the rubric and apply it automatically to relevant assignments or exams.

Table 10. Data Analysis and Reporting User Story

User Story ID	US-8
User Story Name	Data Analysis and Reporting
Actors	Professor
User Story	As a professor, I want to view detailed analytical reports and export them so that I can evaluate course outcomes and share insights with academic departments.
Acceptance Criteria	Given that a professor accesses the analytics dashboard, when they select report parameters such as course, semester, or date range, then the system should generate comprehensive, exportable reports in PDF or Excel format.

Table 11. System Administration User Story

User Story ID	US-9
User Story Name	System Administration
Actors	Administrator
User Story	As a system administrator, I want to manage user accounts and system settings so that I can ensure secure and reliable platform operation.
Acceptance Criteria	Given that an administrator logs into the system, when they create, update, or deactivate user accounts, then the system should apply changes immediately and record all actions in the audit logs.

Table 12. Student Feedback and Notification User Story

User Story ID	US-10
User Story Name	Student Feedback and Notification
Actors	Student, Professor
User Story	As a student, I want to receive immediate feedback and notifications when my grades are published so that I can track my progress and improve performance.
Acceptance Criteria	Given that a professor finalizes grading results, when grades are released, then the system should notify the respective students through dashboard alerts or email and display feedback securely in their accounts.

3.6 Use Case Diagram

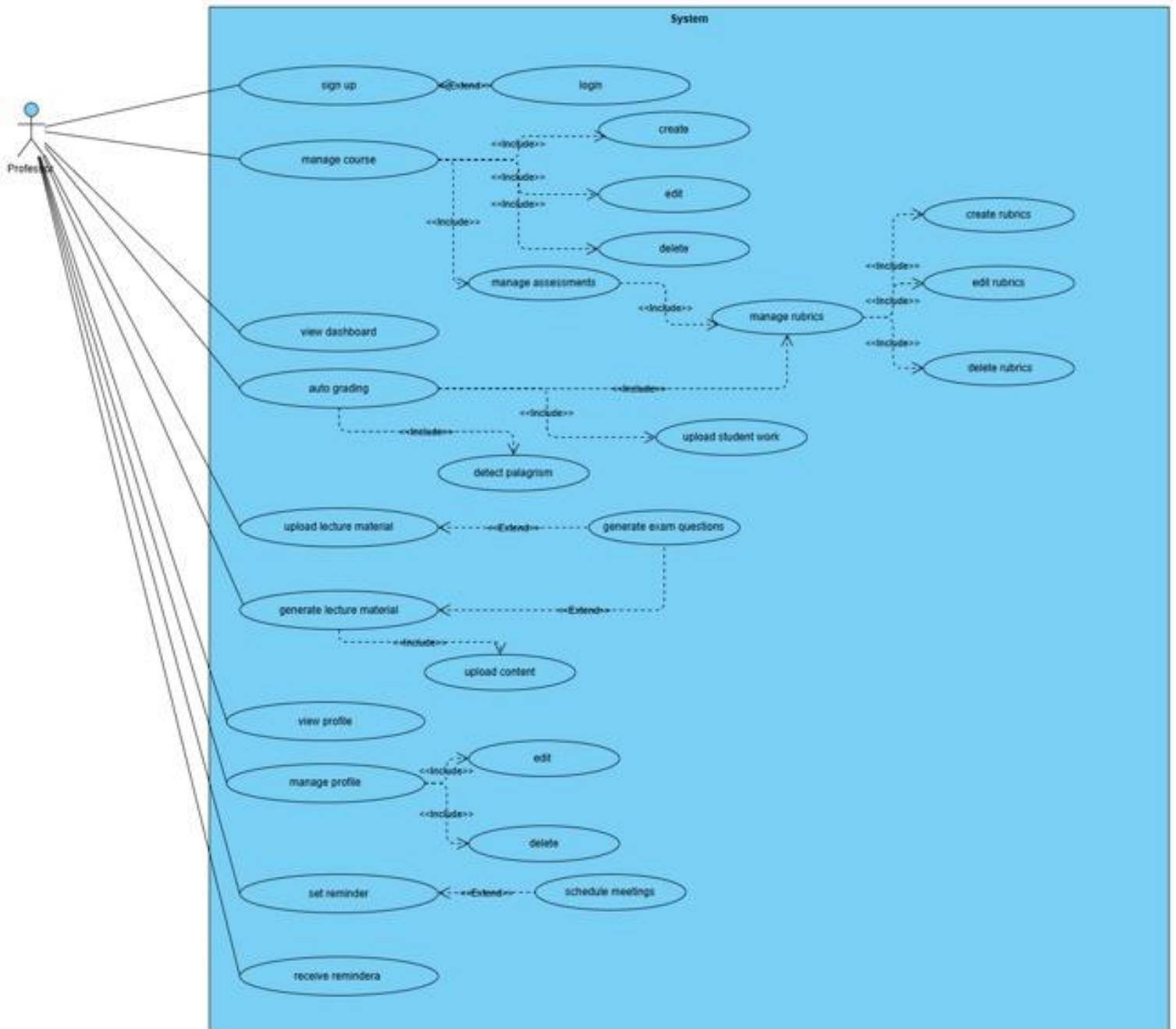


Figure 4. Use Case Diagram – Professor

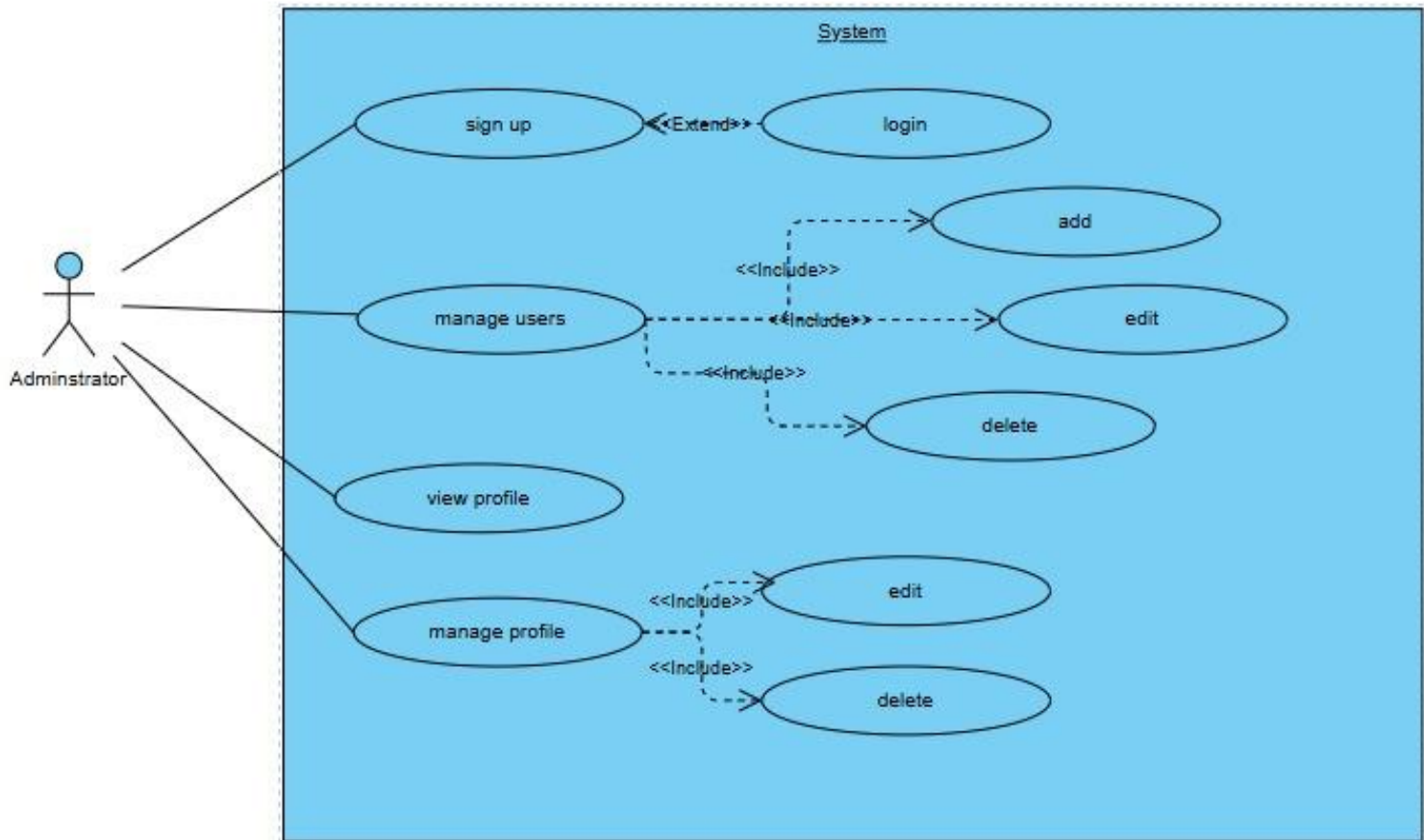


Figure 5. Use Case Diagram – Administrator

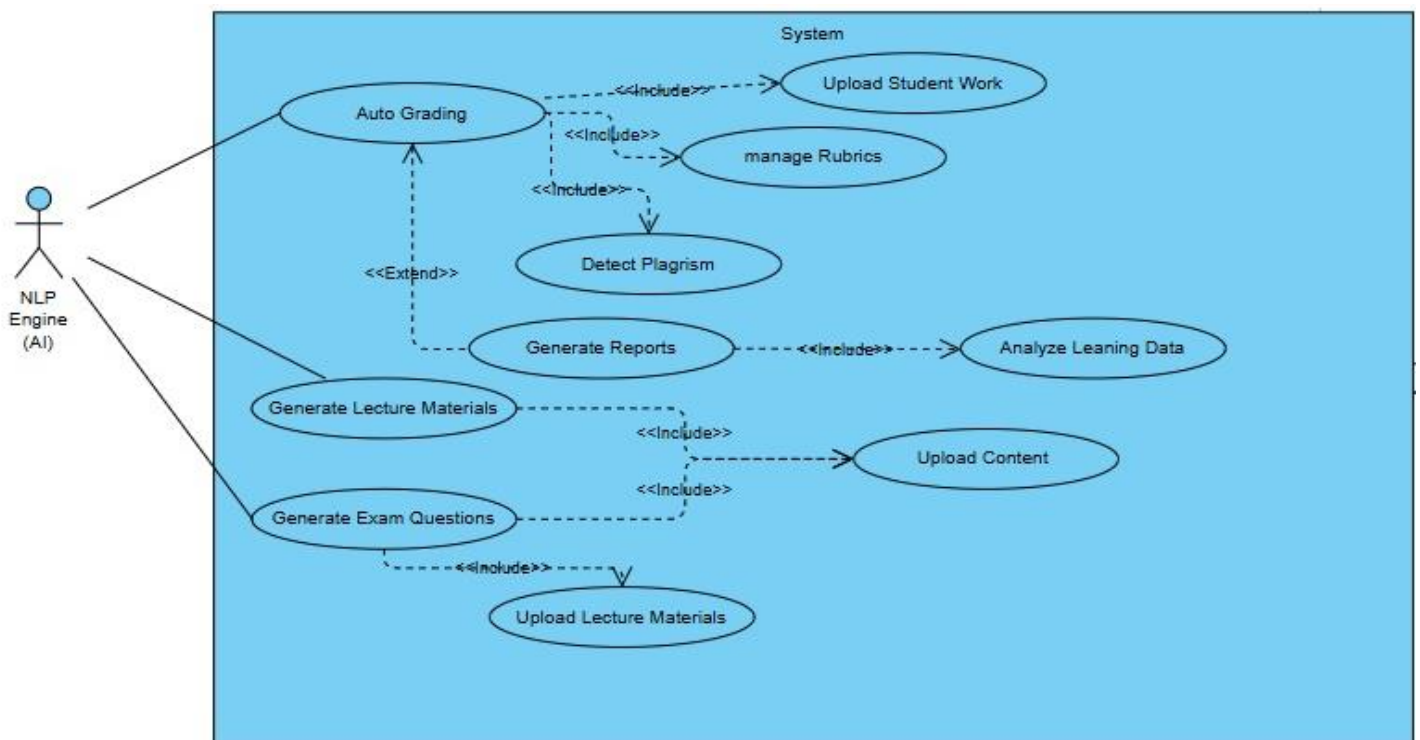


Figure 6. Use Case Diagram – NLP Engine (AI)

3.7 Use Case Description

Table 13. User Signup Use Case

Use Case ID	UCD-1	
Use Case Scenario	User Signup	
Brief Description	The user creates a new account by providing personal details, choosing a password, and verifying their email address.	
Triggering Event	The user wants to access the system for the first time.	
Actors	Professor, Administrator	
Preconditions	- The user must not already have an existing account. - The user must have access to a valid email address.	
Postconditions	-The user's account is successfully created. - A verification email is sent to confirm the registration. - User profile is initialized with basic details.	
Flow of Events	User Action	System Response
	1. The user selects "Sign Up."	1.1. The system displays a registration page.
	2. The user enters name, email, and password.	2.1. The system validates input for completeness and correctness.
	3. The user submits Signup details.	3.1. The system creates the user account. 3.2. A verification email is sent to the provided address. 3.3. The system displays a confirmation message.
Exception Conditions	Invalid email, weak password, or existing account prevents registration.	

Table 14. User Login Use Case

Use Case ID	UCD-2	
Use Case Scenario	User Login	
Brief Description	The user logs into the system by entering their email and password to access their dashboard.	
Triggering Event	The user wants to access their account.	
Actors	Professor, Administrator	
Preconditions	- User must already have a registered account.	
Postconditions	- User successfully logs in and is redirected to the dashboard. - Invalid credentials trigger an error message.	
Flow of Events	User Action	System Response
	1. The user selects "Login."	1.1. The system displays the login page.
	2. The user enters email and password.	2.1. The system validates credentials.
	3. The user submits login details.	3.1. The system verifies the account and grants access. 3.2. If login fails, an error message is shown.
Exception Conditions	Invalid credentials or deactivated account prevent login.	

Table 15. Upload Assignment for Auto-Grading Use Case

Use Case ID	UCD-3	
Use Case Scenario	Upload Assignment for Auto-Grading	
Brief Description	The professor uploads student assignments for automatic grading using the NLP model.	
Triggering Event	The professor wants to evaluate assignments automatically.	
Actors	Professor, NLP Engine (AI)	
Preconditions	- The professor must be logged in. - The uploaded file must be valid and in a supported format.	
Postconditions	- Assignments are analyzed and graded automatically. - Feedback and grades are displayed on the dashboard.	
Flow of Events	User Action	System Response
	1. The professor selects "Upload Assignment."	1.1. The system displays an upload form.
	2. The professor uploads files and sets grading criteria.	2.1. The system validates file type and rubrics.
	3. The professor confirms upload.	3.1. NLP model analyzes content and generates grades. 3.2. Results are displayed and saved.
Exception Conditions	Invalid file type, corrupted data, or system error prevents grading.	

Table 16. Detect Plagiarism Use Case

Use Case ID	UCD-5	
Use Case Scenario	Detect Plagiarism	
Brief Description	The professor checks student submissions for plagiarism using online and institutional databases.	
Triggering Event	The professor wants to verify academic integrity.	
Actors	Professor, NLP Engine (AI)	
Preconditions	- Professor must be logged in. - Assignments are uploaded.	
Postconditions	- System generates a plagiarism report showing similarity percentage and matched sources.	
Flow of Events	User Action	System Response
	1. The professor selects "Check Plagiarism."	1.1. System retrieves uploaded assignments.
	2. The professor confirms files for scanning.	2.1. System compares content with databases and generates report.
	3. The professor views and downloads the report.	3.1. System saves report for future access.
Exception Conditions	Internet or database connection failure stops the scan.	

Table 17. Generate Exam Questions Use Case

Use Case ID	UCD-6	
Use Case Scenario	Generate Exam Questions	
Brief Description	The professor generates exam questions using AI model	
Triggering Event	The professor wants to create an exam.	
Actors	Professor, NLP Engine (AI)	
Preconditions	Professor must be logged in. - Lecture content or notes must be uploaded.	
Postconditions	- Questions are generated, editable, and exportable.	
Flow of Events	User Action	System Response
	1. The professor selects "Generate Exam Questions."	1.1. System requests course content.
	2. The professor uploads lecture notes.	2.1. AI generates questions in multiple formats.
	3. The professor reviews and exports questions.	3.1. System displays editable output.
Exception Conditions	Insufficient or unclear content prevents question generation.	

Table 18. Analyze Data Use Case

Use Case ID	UCD-7	
Use Case Scenario	Analyze Data	
Brief Description	The professor uploads sheets (grades, attendance) to generate performance insights.	
Triggering Event	The professor wants to analyze student performance.	
Actors	Professor	
Preconditions	-Professor must be logged in. - file follows the required format.	
Postconditions	- Analytics and charts are displayed on the dashboard.	
Flow of Events	User Action	System Response
	1. The professor selects "Analyze Data."	1.1. System prompts for file upload.
	2. The professor uploads the file.	2.1. System validates structure and content.
	3. The professor views analysis.	3.1. System displays visual analytics.
Exception Conditions	Invalid format → Error message displayed.	

Table 19. Manage Professor Profile Use Case

Use Case ID	UCD-8	
Use Case Scenario	Manage Professor Profile	
Brief Description	The professor manages their personal information, research, and academic history	
Triggering Event	The professor wants to update or view their profile.	
Actors	Professor	
Preconditions	Professor must be logged in.	
Postconditions	- Profile data is saved and displayed successfully.	
Flow of Events	User Action	System Response
	1. The professor selects "Profile."	1.1. System displays profile page.
	2. The professor selects edit details.	2.1. System displays editing info page.
	3. The professor enters changes and click save.	3.1. System validates inputs. 3.2. System updates profile and confirms success.
Exception Conditions	Missing or invalid fields trigger a warning.	

Table 20. Manage Courses Use Case

Use Case ID	UCD-9	
Use Case Scenario	Manage Courses	
Brief Description	This use case allows the professor to create, edit, and delete courses within the Profound system. Each course contains information such as title, description, code, schedule, and linked assessments. It also includes creating and assigning assessments and analyzing course learning data.	
Triggering Event	The professor wants to add a new course, update course information, or remove an existing course.	
Actors	Professor	
Preconditions	- The professor must be logged in.	
Postconditions	A new course is created, updated, or deleted successfully. - Any linked assessments and analytics are automatically updated.	
Flow of Events	User Action	System Response
	1. The professor selects "Manage Courses."	1.1. The system displays the list of existing courses.
	2. The professor chooses to add, edit, or delete a course.	2.1. The system provides corresponding forms for data entry or confirmation.
	3. The professor fills in course details (title, code, description, schedule, etc.).	3.1. The system validates input for completeness and uniqueness.
	4. The professor submits the form.	4.1. The system saves or updates course information in the database.
Exception Conditions	- Invalid or missing course data prevents submission.	

Table 21. Generate Lecture Materials Use Case

Use Case ID	UCD-10	
Use Case Scenario	Generate Lecture Materials	
Brief Description	This use case enables the professor to upload a topic or file, and the system automatically generates lecture materials such as slides, summaries, and key points using the AI (NLP Engine).	
Triggering Event	The professor wants to prepare lecture materials quickly and efficiently using AI.	
Actors	Professor, NLP Engine (AI)	
Preconditions	- The professor must be logged in. - A valid topic or source document must be provided.	
Postconditions	- Lecture materials are generated and displayed for review. - The professor can download or edit the materials.	
Flow of Events	User Action	System Response
	1. The professor selects "Generate Lecture Materials."	1.1. The system displays options to input a topic or upload a reference file.
	2. The professor provides the topic or uploads related content.	2.1. The system validates the input and sends it to the NLP Engine.
	3. The professor submits the request.	3.1. The AI model processes the input and generates lecture notes, slides, and key points.

	4. The professor reviews and optionally edits the output.	4.1. The system allows editing and saving of the AI-generated materials.
Exception Conditions	- Missing or unclear input topic. - AI service unavailable or error during generation.	

Table 22. Manage Rubrics Use Case

Use Case ID	UCD-11	
Use Case Scenario	Manage Rubrics	
Brief Description	This use case allows the professor to create, edit, and delete grading rubrics for assignments or exams. Rubrics define the evaluation criteria, weightage, and scoring levels used for both manual and automated grading.	
Triggering Event	The professor wants to create or modify a grading rubric for an assessment.	
Actors	Professor	
Preconditions	- The professor must be logged in. - The professor must have permission to manage course assessments.	
Postconditions	- A new rubric is created, updated, or deleted successfully. - The rubric is linked to the relevant assessment for grading.	
Flow of Events	User Action	System Response
	1. The professor selects "Manage Rubrics."	1.1. The system displays the list of existing rubrics and their details.

	2. The professor chooses to add, edit, or delete a rubric.	2.1. The system opens the rubric management interface.
	3. The professor defines or updates rubric items such as criteria, score range, and weight.	3.1. The system validates all input data for completeness and logical scoring.
	4. The professor saves the rubric.	4.1. The system stores the rubric and associates it with a specific course or assessment.
	5. The professor can later return to modify or remove a rubric.	5.1. The system updates or deletes the rubric record in the database.
Exception Conditions	-Missing or invalid rubric fields prevent saving. -Attempting to delete rubric linked to active assessments triggers a confirmation warning.	

3.8 Class Diagram

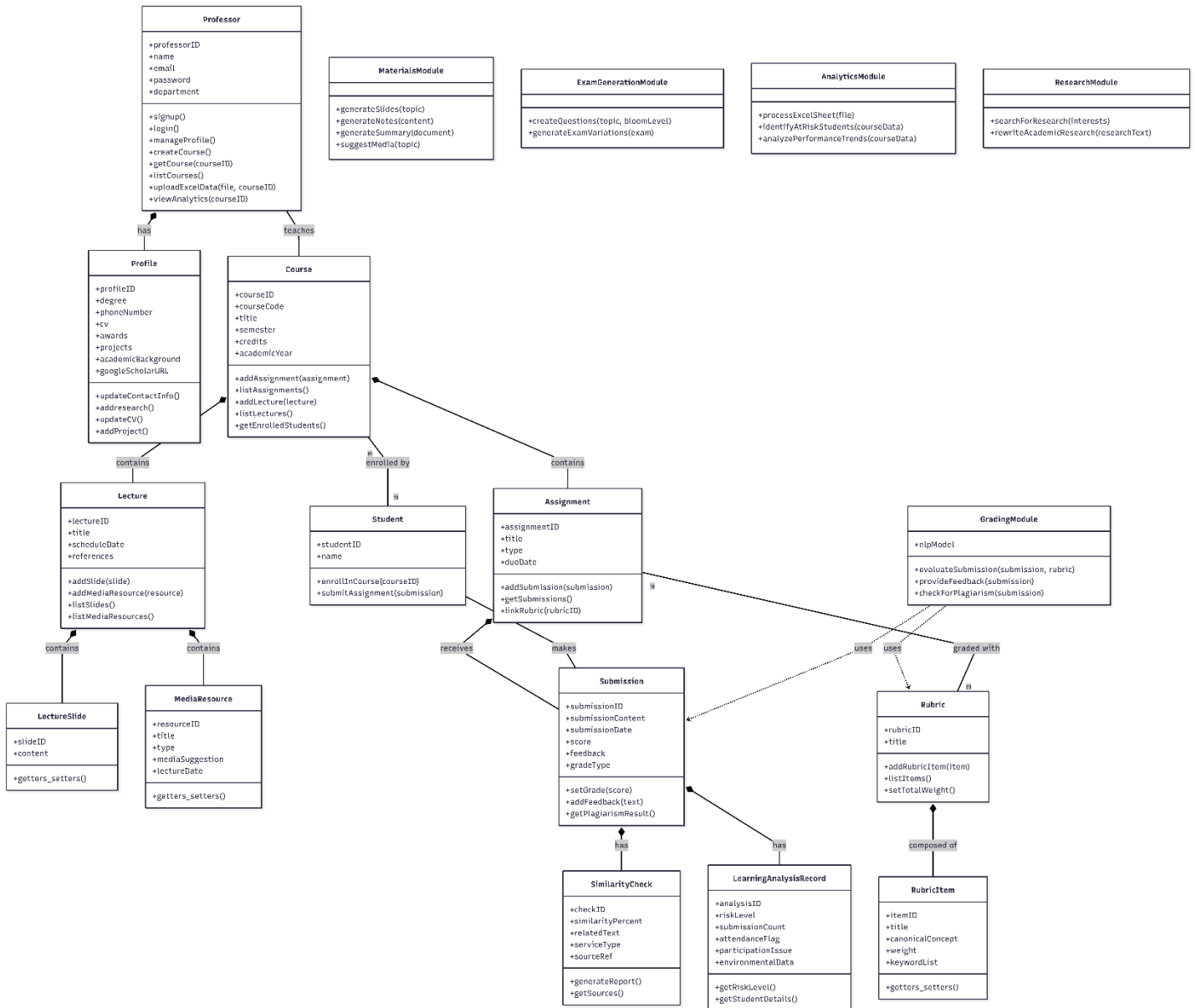


Figure 7. Class Diagram

3.9 ERD Model

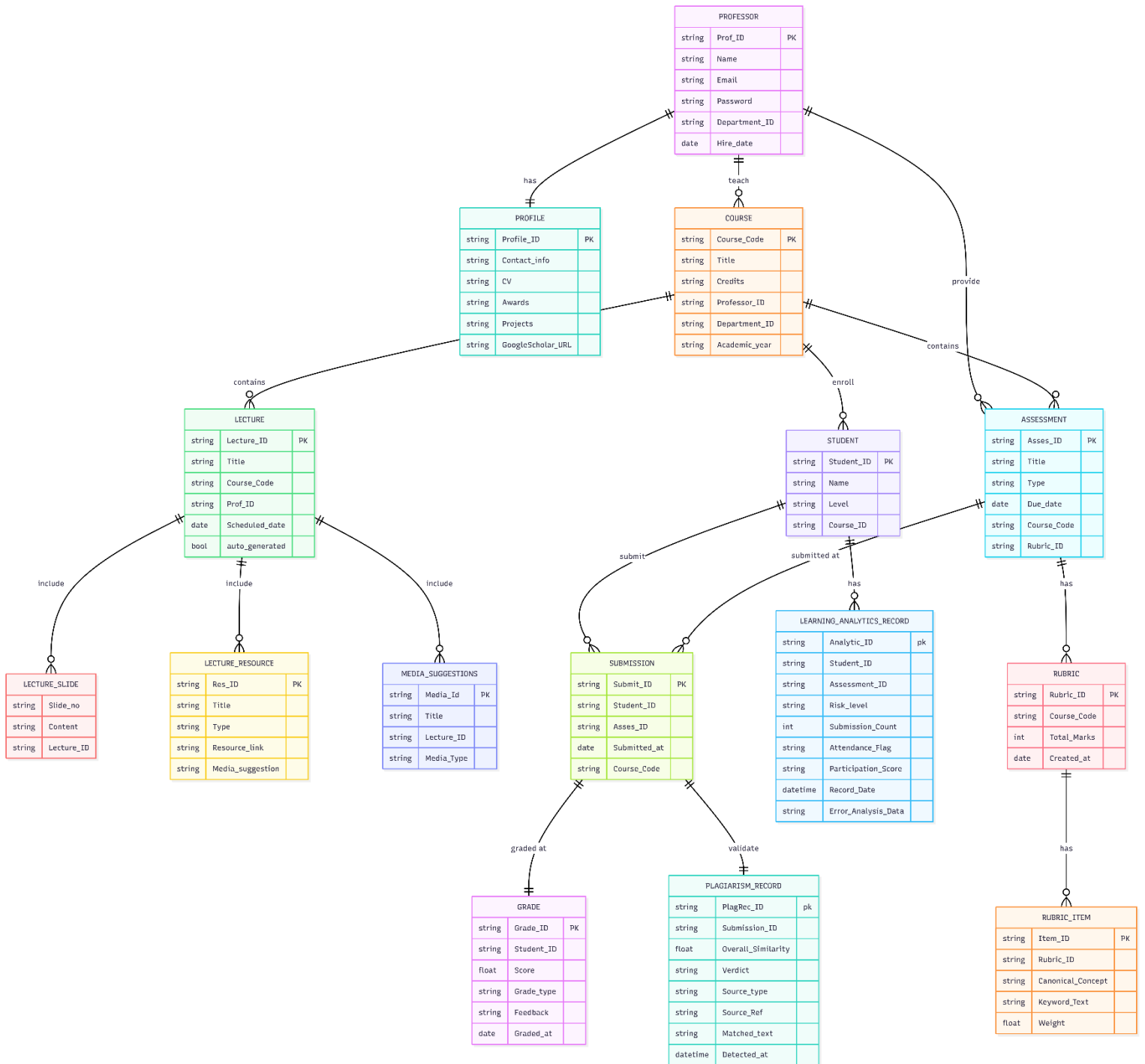


Figure 8. ERD Model

4 Profound Design and Implementation

4.1 Profound UI&UX

Prototyping Tool: Figma We utilized **Figma** as our primary tool for developing the system's prototype. It was selected due to its cloud-based collaborative environment, which allowed the team to work simultaneously on the design in real-time. Figma's robust prototyping features enabled us to create high-fidelity, interactive mockups that accurately simulate the user journey, ensuring a clear vision before the development phase began.

Project Link:

[Profound](#)

4.1.1 User Interface

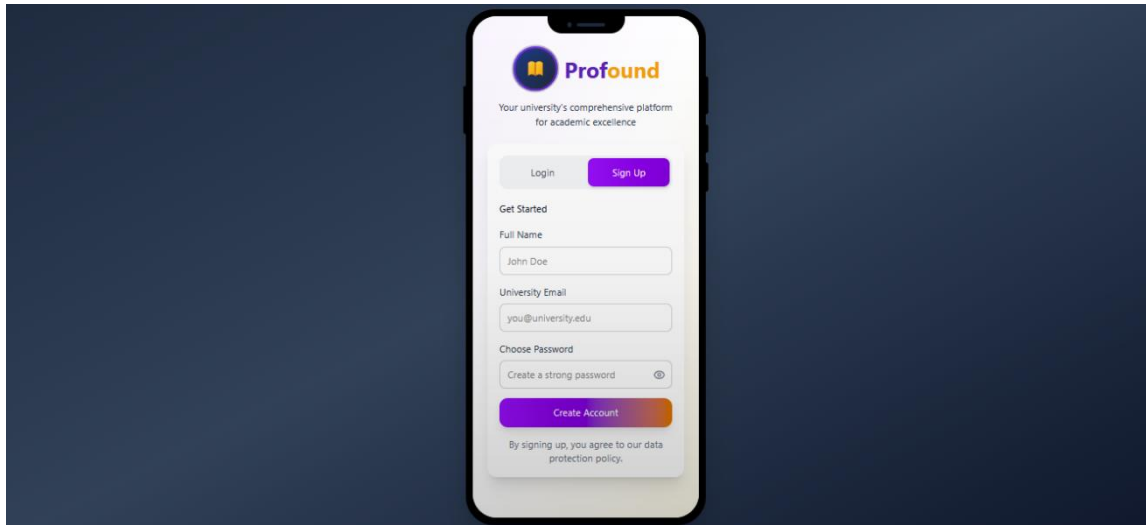


Figure 9. UI User - Sign Up

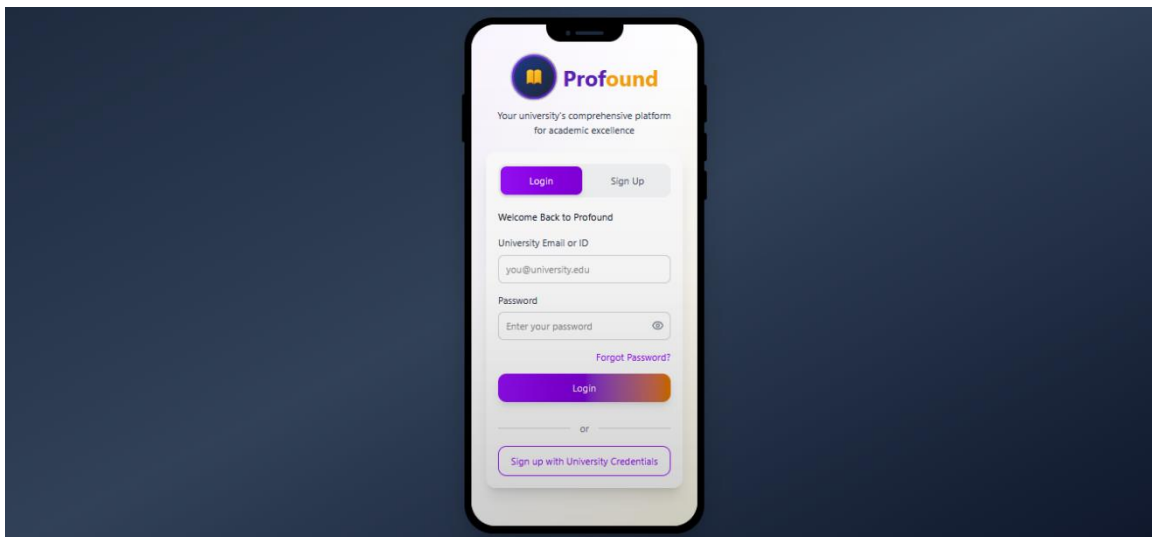


Figure 10. UI User - Login

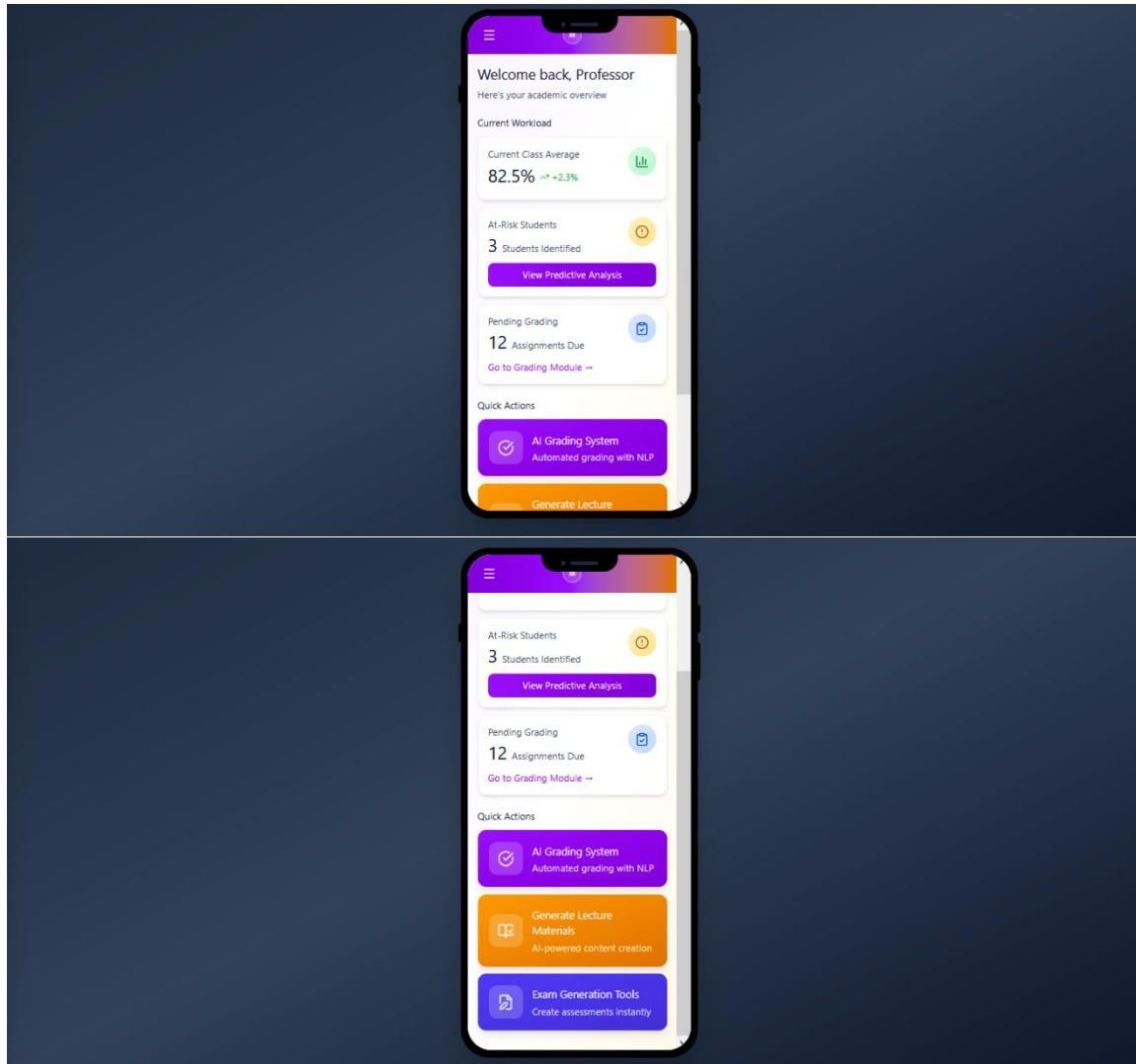


Figure 11. UI User – Dashboard Page

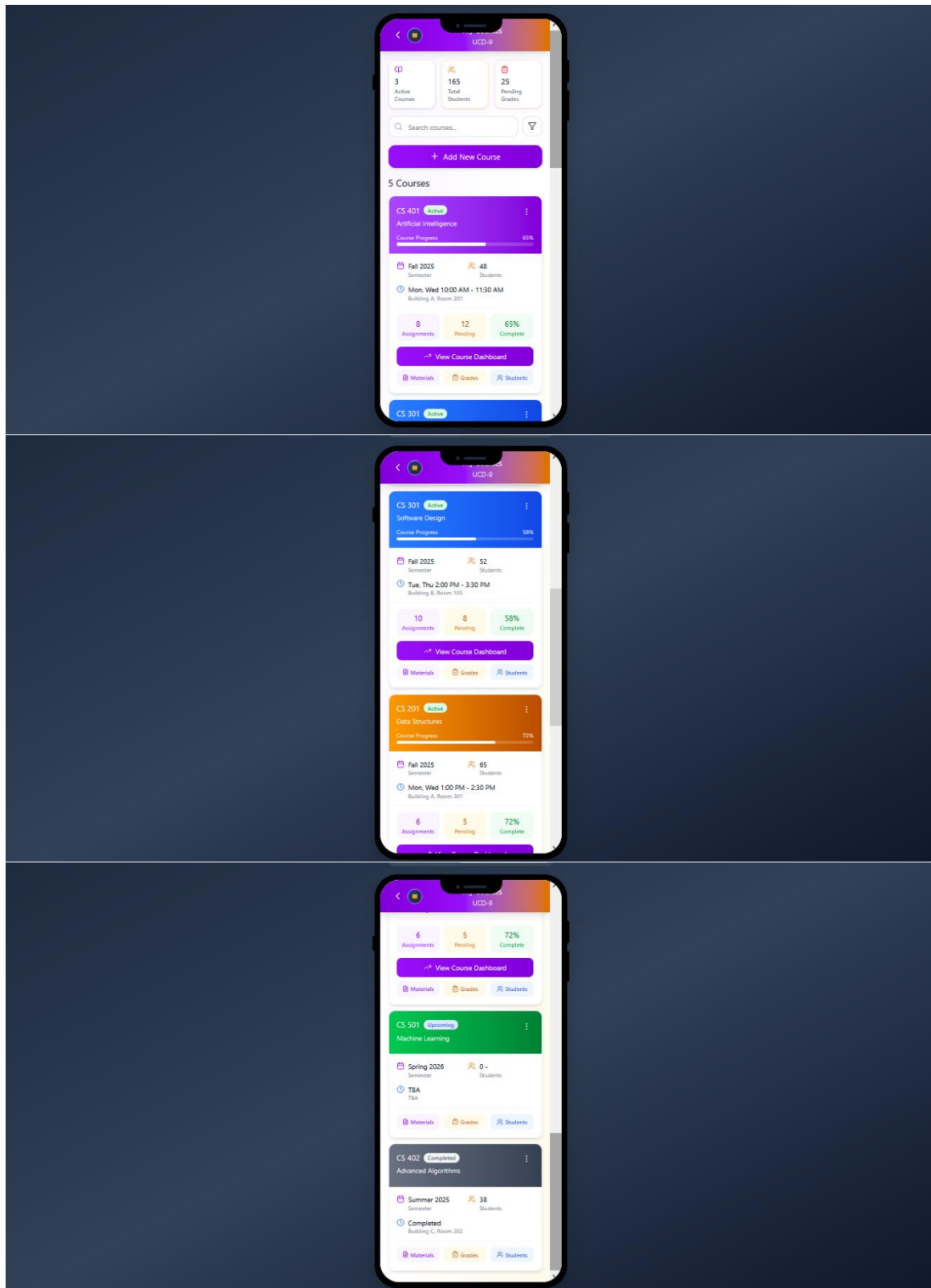


Figure 12. UI User – Courses Page

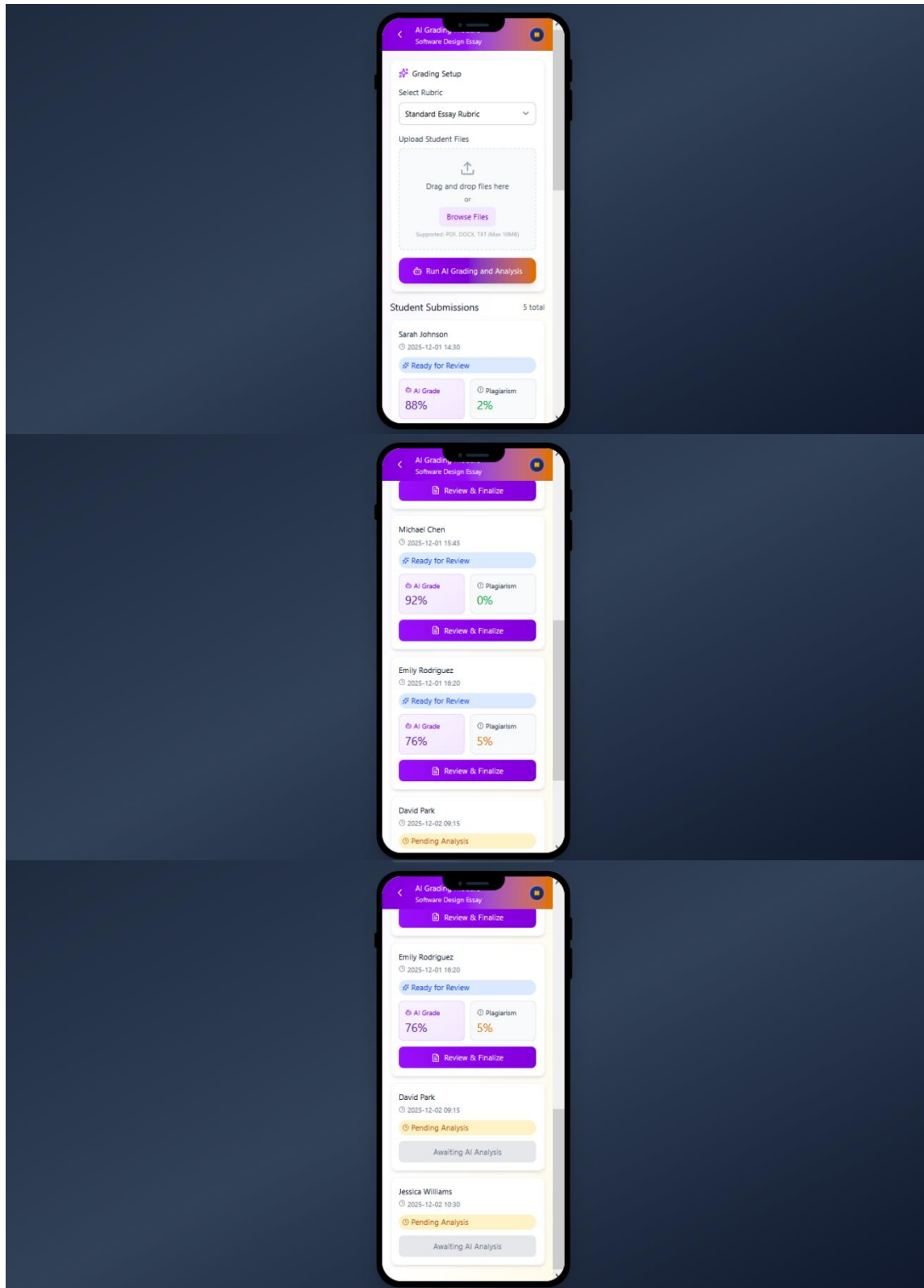
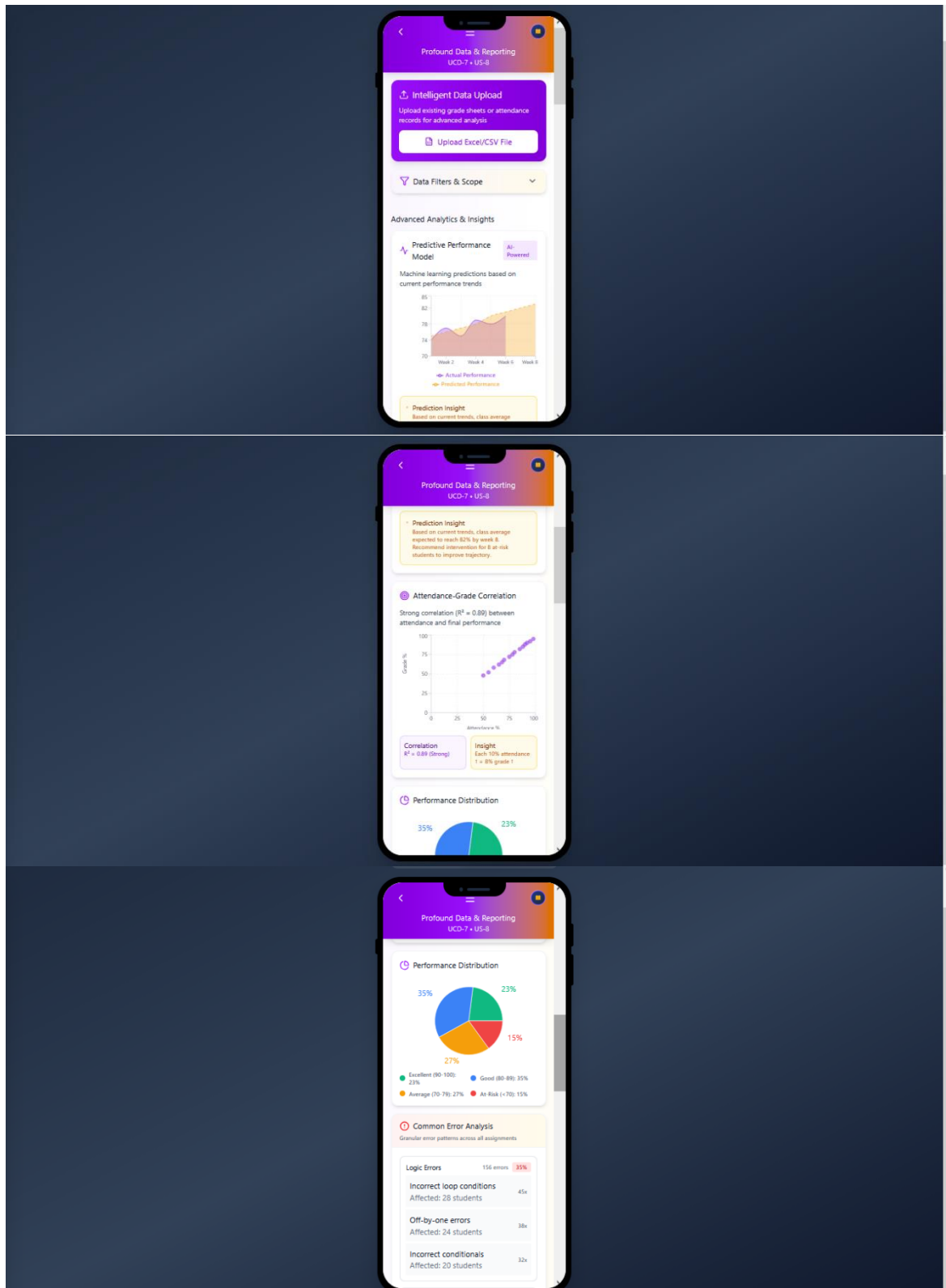


Figure 13. UI User – Grading Page



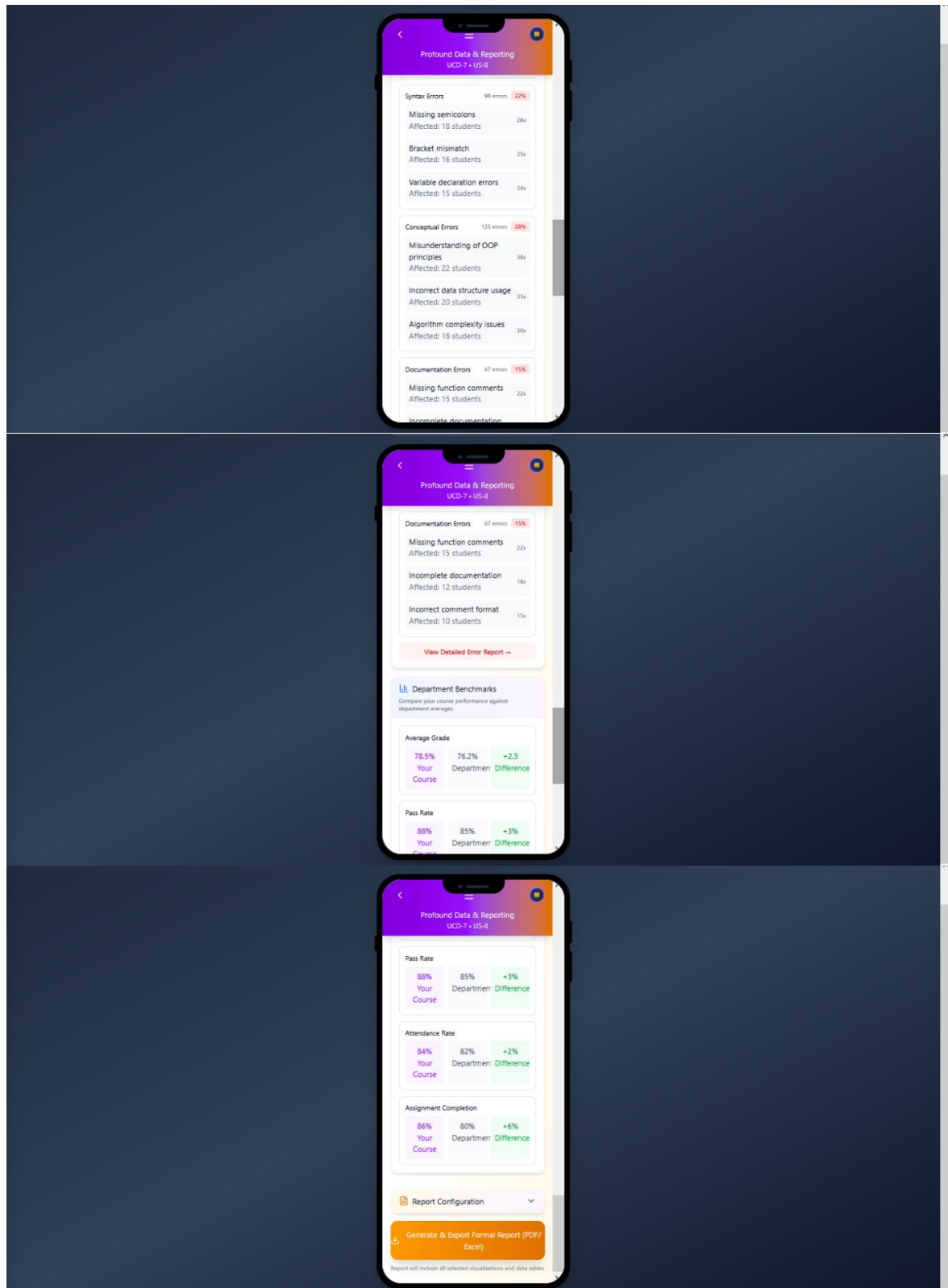
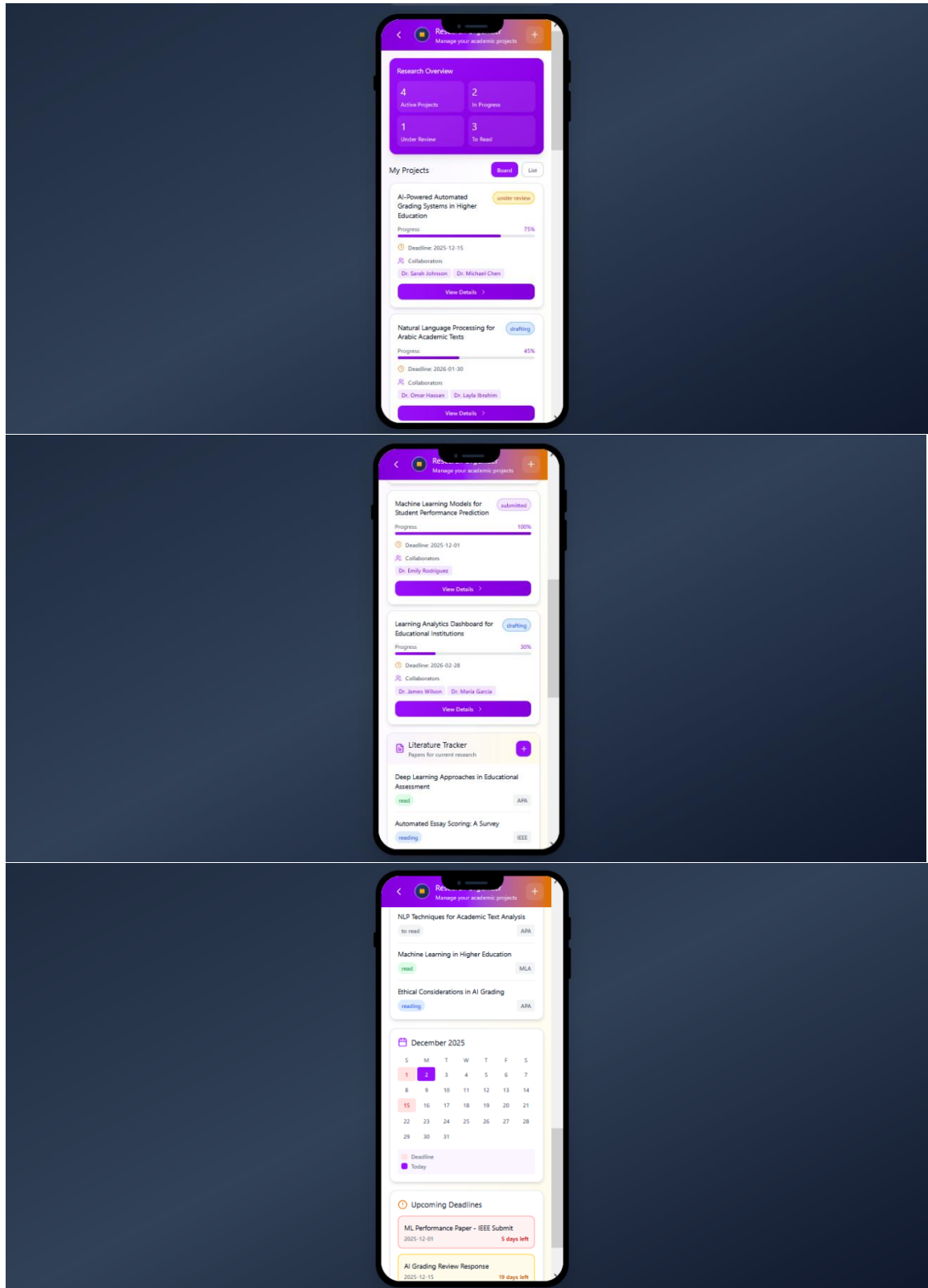


Figure 14. UI User – Analytics and Report Page



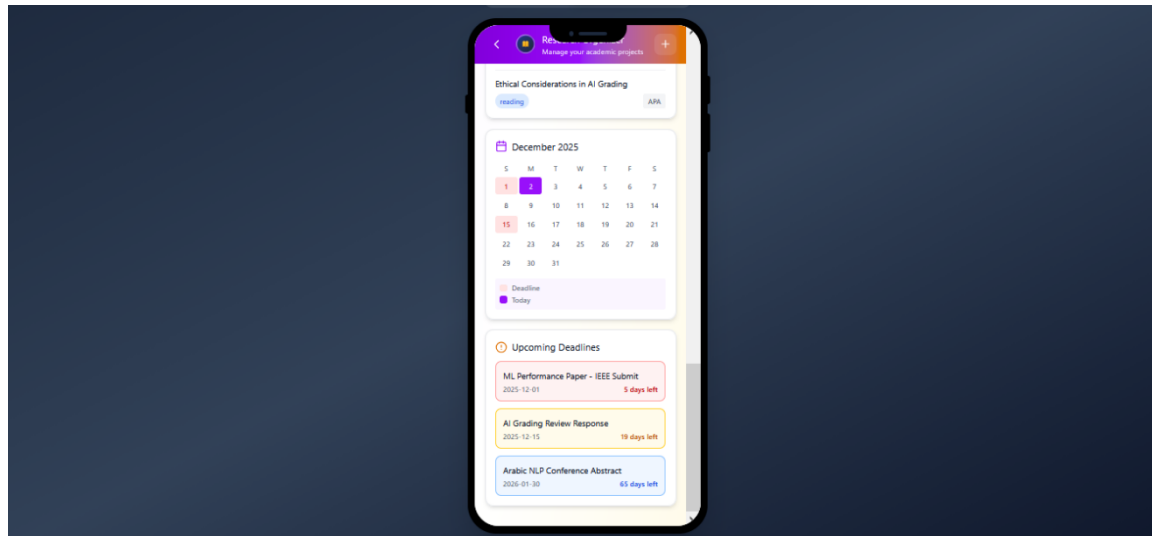


Figure 15. UI User – Research Organizer Page

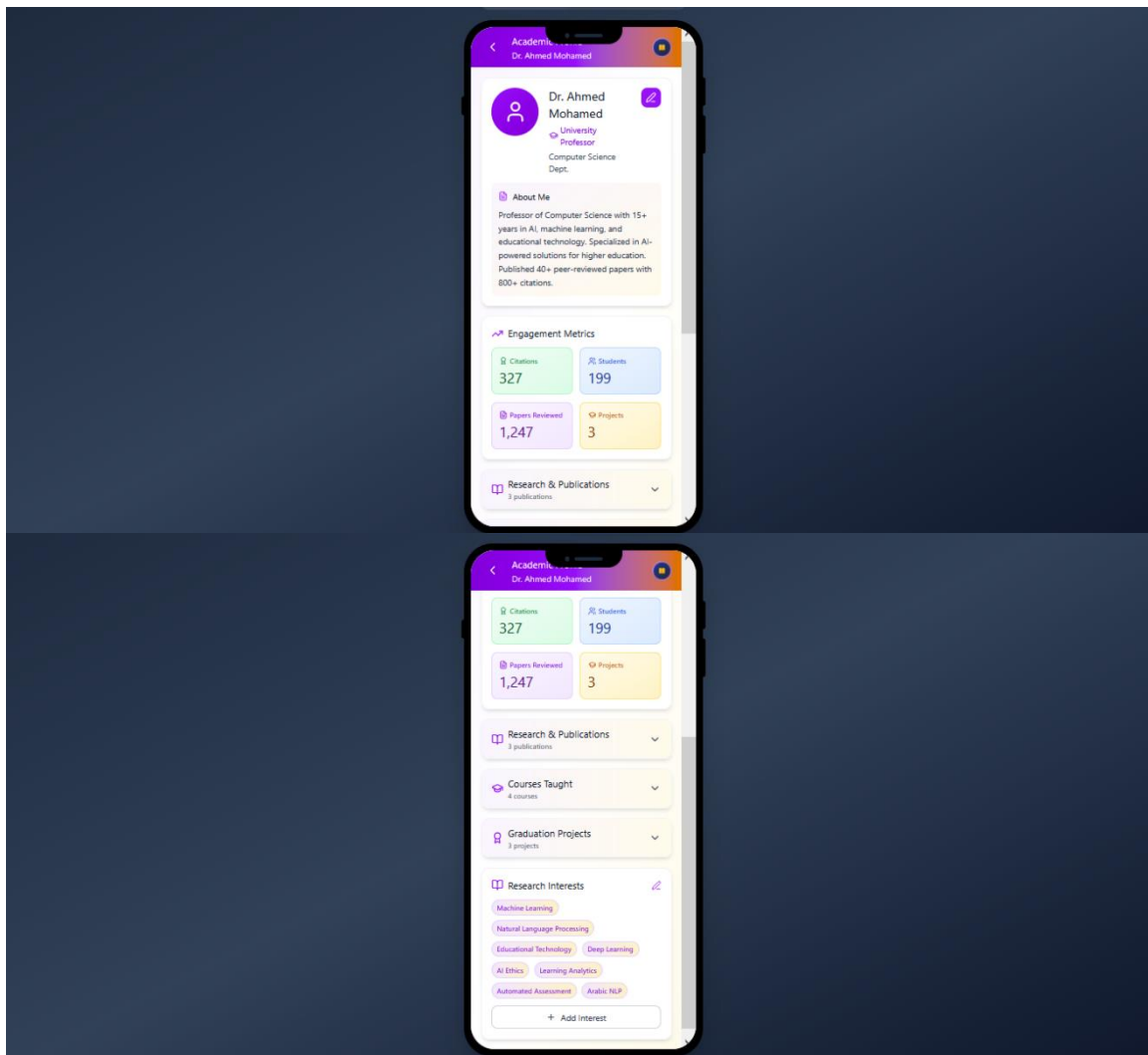


Figure 16. UI User – Academic Profile Page

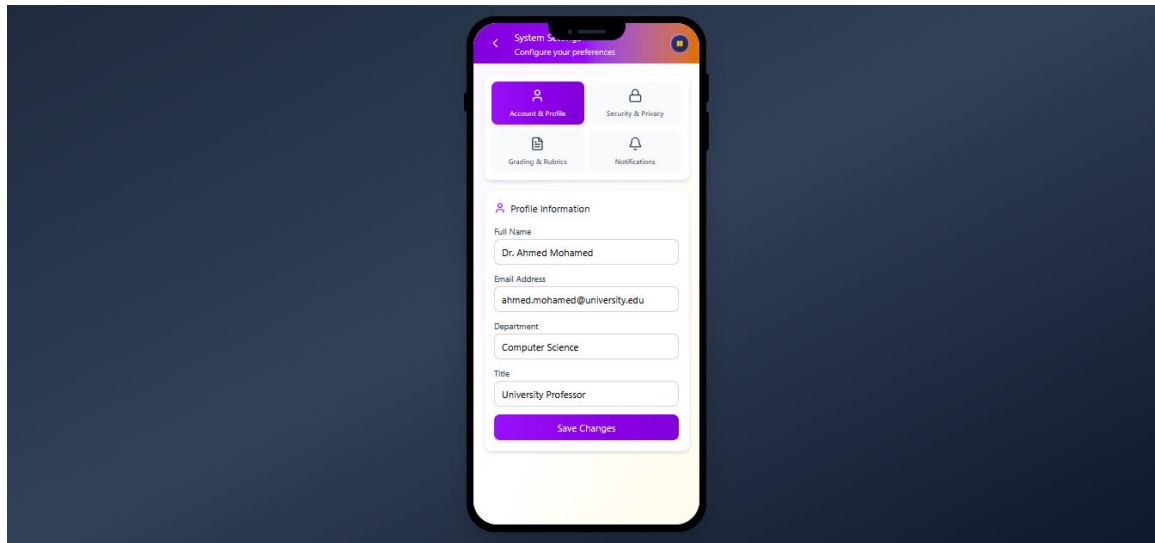


Figure 17. UI User – Settings – Account & Profile Page

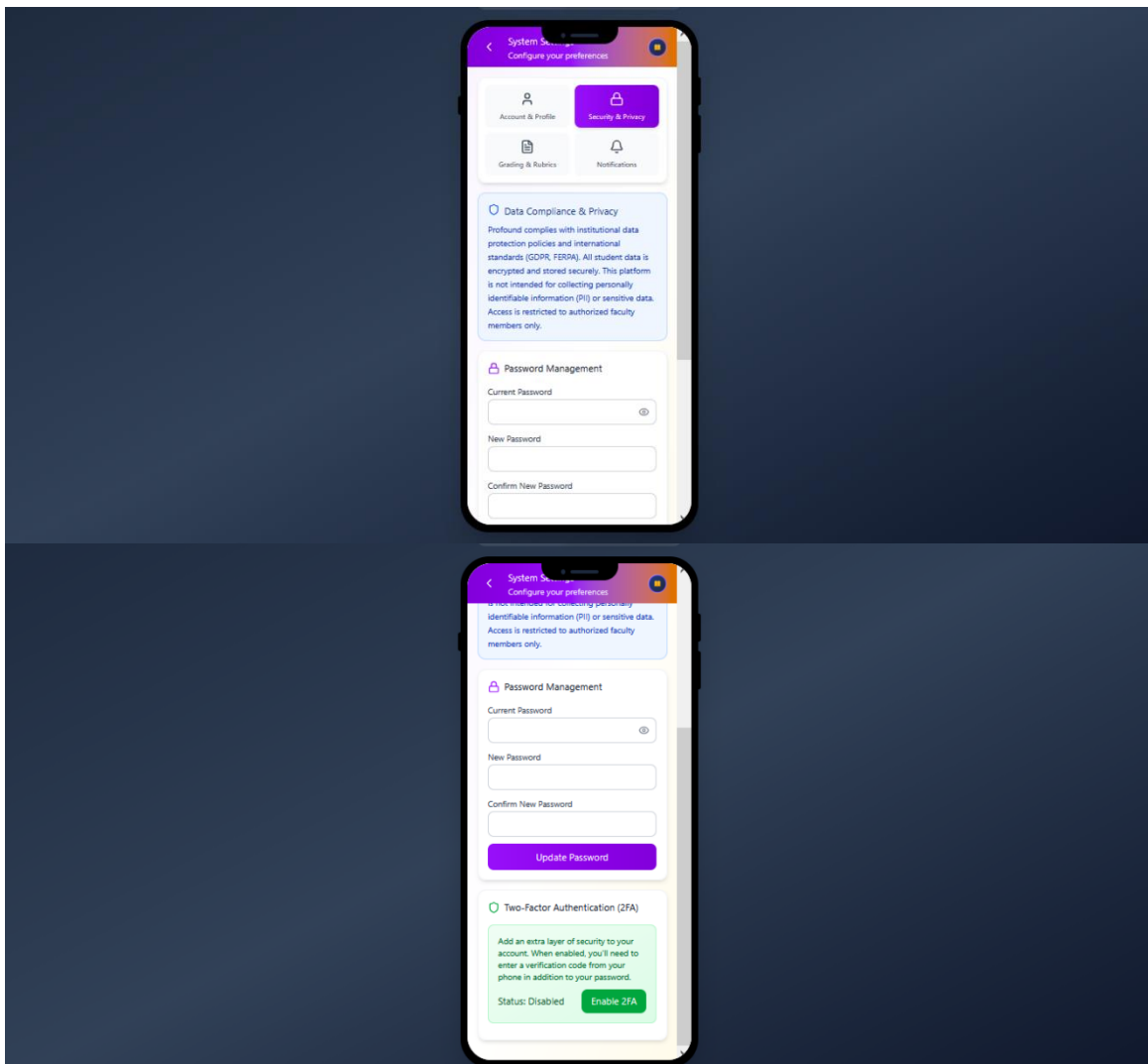


Figure 18. UI User – Settings – Security & Privacy Page

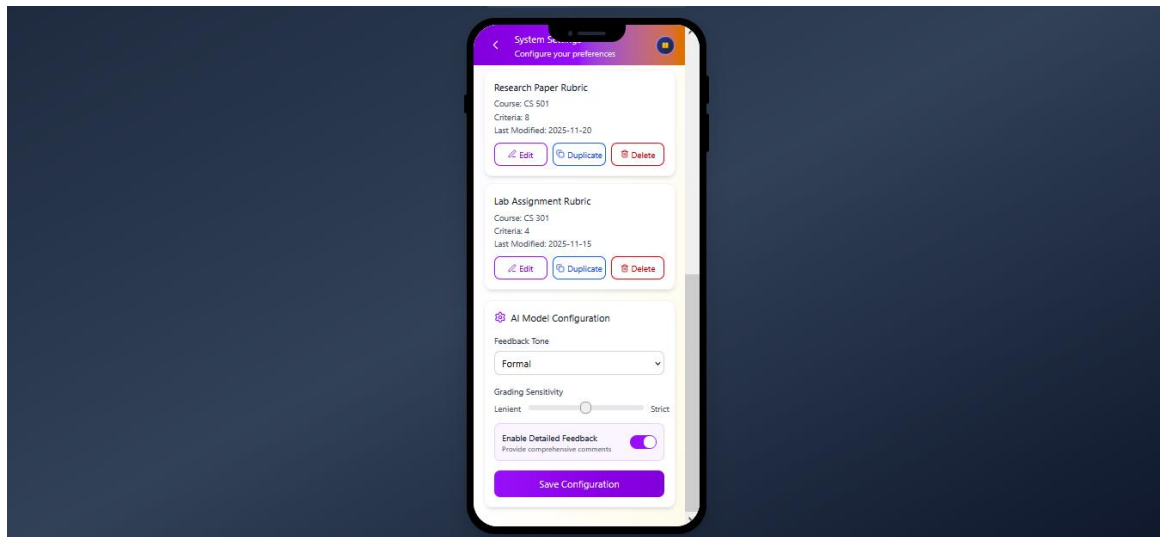
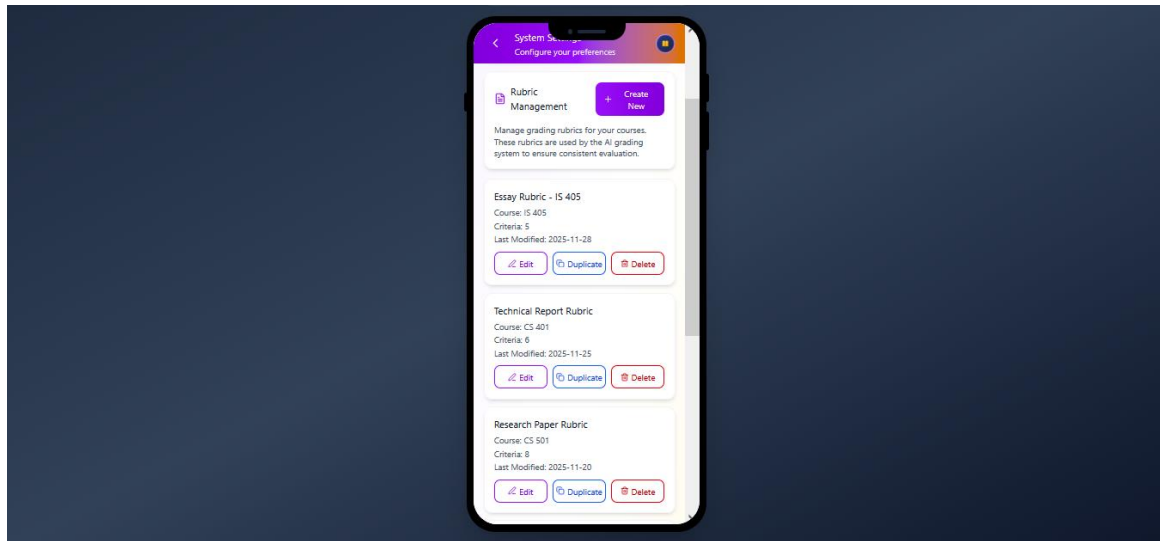


Figure 19. UI User – Settings – Grading & Rubrics Page

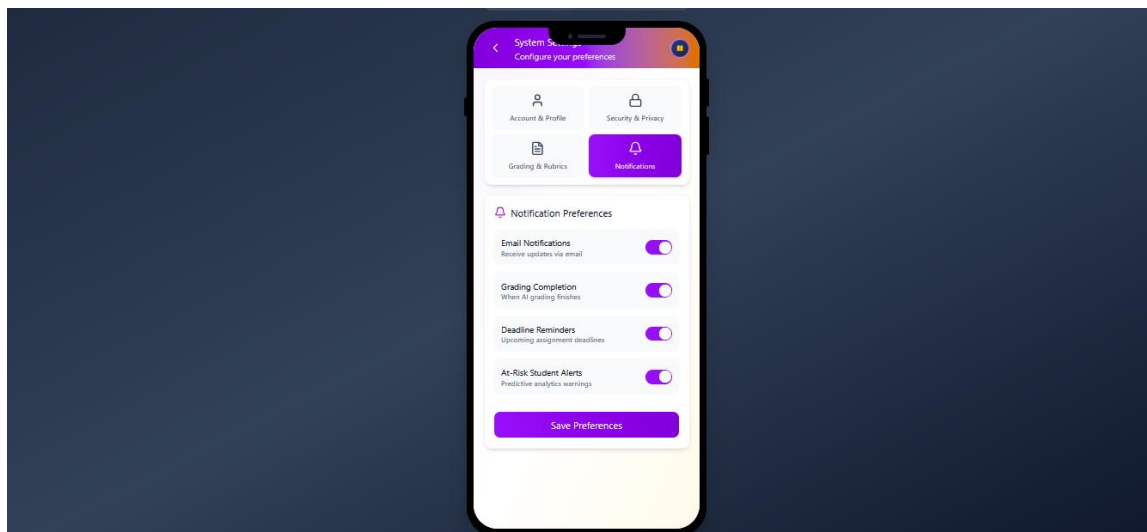


Figure 20. UI User – Settings – Notification Page